

THE INTERGENERATIONAL EFFECTS OF EARLY-LIFE POLLUTION EXPOSURE.*

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Abstract

Using newly linked survey and administrative data, we provide early evidence that regulation-induced improvements in prenatal air quality have intergenerational effects on human capital accumulation in the United States. Supporting evidence suggests that intergenerational transmission arises more from greater parental resources and investments than from heritable channels. Overall, our findings suggest that within-generational analyses of marginal damages could substantially underestimate the total benefits of improving environmental quality, and highlight the importance of directly evaluating intergenerational effects.

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1 Introduction

It is well established that disparities in health and economic outcomes persist across generations. It is also widely understood that exposure to air pollution and other environmental risks during the in utero period plays a critical role in shaping economic opportunity, with persistent effects on health, productivity, and later-life earnings (Chay and Greenstone, 2003a,b; Currie et al., 2009; Isen et al., 2017; Colmer et al., 2021). Less clear is how much differences in exposure to environmental risks shape the outcomes of the next generation, contributing to the intergenerational persistence of health and economic outcomes. While it is plausible that shocks to the early-life environment might have intergenerational consequences, data limitations have impeded our ability to directly evaluate how much they shape the economic outcomes of future generations.

We construct a new linked dataset containing more than 150 million parent–child links. Combining this data to respondents in the American Community Survey, we present the first evidence that early-life pollution exposure affects not only those who were directly exposed, but also the economic opportunities of their children, decades later. We do this by leveraging the introduction of the 1970 Clean Air Act Amendments (CAAA), which imposed county-level limits on the maximum allowable concentrations of total suspended particles (TSP), resulting in substantial improvements in ambient air quality in hundreds of counties throughout the United States (Chay and Greenstone, 2003a; Isen et al., 2017). Children conceived in the year before improvements in air quality experienced one more year of pollution than children conceived after these improvements. For counties that were already in compliance, there were no regulation-induced changes in air quality. This difference-in-differences research design plausibly identifies the effects of an additional year of clean air at this critical stage of the life cycle because there was no persistent change in air quality between cohorts in areas that experienced reductions in TSP. Individuals born in the year before improvements did not experience the benefits of reduced air pollution during the first year of life, but they did benefit after the regulations went into effect. We evaluate whether

these acute improvements in air quality differentially affected the later-life outcomes of the next generation, i.e., the *children* of the cohorts who directly experienced this additional year of better air quality, compared to the children of cohorts who did not experience an additional year of better air quality.

We estimate that first-generation individuals who experienced lower exposure to pollution in the gestational environment are more likely to have children who end up attending college 40-50 years later. A $10\mu g/m^3$ reduction in first-generation gestational TSP exposure is associated with a 1.66–1.76 percentage point increase in the likelihood of second-generation college attendance.¹ This is equivalent in magnitude to the intergenerational effects of 0.098–0.121 Head Start programs (Barr and Gibbs, 2022). We also present early suggestive evidence that reductions in first-generation prenatal pollution exposure may translate into higher second-generation earnings and employment.

While the reductions in pollution induced by the 1970 Clean Air Act delivered health and economic benefits for those directly affected (Chay and Greenstone, 2003a; Chay et al., 2003; Sanders and Stoecker, 2015; Isen et al., 2017), our evidence suggests that within-generation analyses that look only at the later life outcomes of those directly affected could substantially underestimate the total benefits of investments in environmental quality. Back-of-the-envelope calculations show that the combined discounted lifetime earnings effect for both the first and second generation is 50-85% larger than the total earnings effect for the first generation alone (Isen et al., 2017). This combined total accounts for 40-60% of the monetized damages associated with infant mortality (Chay and Greenstone, 2003a). To date, the mortality effects of pollution have been the largest monetized damage in formal benefit–cost analysis of environmental regulations. Mortality benefits currently account for

¹The implied estimate from the reduced-form effect of the 1970 CAAA, which, on average, induced an $11.55\mu g/m^3$ reduction for the parents of the second-generation sample, is almost identical. An $11.55\mu g/m^3$ reduction in first-generation gestational TSP exposure is associated with a 1.92–2.03 percentage point increase in the likelihood of second-generation college attendance. Our reduced-form estimates range from 1.92–2.08 percentage points. In proportional terms, an $11.5\mu g/m^3$ decrease in TSP exposure from the 1971 baseline would be similar to the cross-sectional difference in the Black–White particulate matter exposure gap in the early 2000s (Currie et al., 2023).

85% of the benefits for the Clean Air Act calculated by the Environmental Protection Agency (EPA) (EPA, 2020). Failing to account for long-run and intergenerational effects likely leads us to substantially understate the total returns to improvements in environmental quality.

In light of our main finding, we also investigate the mechanisms that drive intergenerational transmission. First, using information on biological and non-biological parent–child links, we provide very suggestive evidence that $\sim 30\%$ of our results could be attributed to heritable transmission, suggesting that parental resources and investments are an important channel through which intergenerational transmission occurs. Second, we estimate that the association between prenatal exposure to pollution and later-life earnings within the first generation could plausibly explain 20% of the estimated college attendance effect on the second generation.² While our effects on first-generation earnings are noisily estimated, much larger effects of pollution on first-generation earnings or effects of parental income on college attendance would be necessary to fully reconcile our results. Third, we find no evidence that differential selection into the second-generation sample, through effects on fertility or family structure, explain our findings.

If reductions in early-life pollution exposure increase health and wealth, parents may be better able to spend time and make investments in their children, improving outcomes for the second generation (Neidell, 2000; Heckman and Carneiro, 2003; Cunha et al., 2010; Del Boca et al., 2013; Heckman et al., 2013; Lundberg, 2017). To explore this, we use linked data from the American Time-Use Survey and provide suggestive evidence that lower first-generation prenatal exposure to TSP is associated with first-generation parents spending more quality time with the second generation when they are children (Price, 2008).

These additional margins through which parents affect child outcomes appear to be important for two reasons. First, as noted, we can only reconcile $\sim 50\%$ of our total second-

²If we take the largest estimate from the existing literature on the effects of parental earnings on college attendance (Lovenheim and Lockwood Reynolds, 2013), assume that both parents experienced reductions in pollution, and assume that our findings are driven entirely by the increase in household earnings, we calculate a 0.33 percentage point increase in the likelihood of attending college (20% of our most conservative estimate). If we use the upper bound of the 95% confidence interval from Lovenheim and Lockwood Reynolds (2013), we can explain 32% of our most conservative estimate.

generation college attendance effect through biological and income-related mechanisms. Second, naive calculations of intergenerational persistence using existing intergenerational elasticities would substantially underestimate the intergenerational effects of early-life pollution exposure.³ We naively calculate that the effect of early-life pollution exposure on the combined first-generation and (implied) second-generation discounted lifetime earnings is up to 25% larger than the total effect on first-generation earnings. Collectively, our findings highlight the importance of directly evaluating the intergenerational consequences of early childhood environments and public policy.

In addition to providing the first evidence that air pollution and environmental regulations have intergenerational consequences, our paper contributes to a nascent literature documenting persistence in the economic consequences of early childhood shocks across generational boundaries (Nilsson, 2017; Black et al., 2019; Heckman and Karapakula, 2019; Barr and Gibbs, 2022; East et al., 2023). Much of the existing literature focuses on health at birth, or on more targeted anti-poverty programs in the United States.⁴ Our findings provide early evidence on the later-life intergenerational effects of an important margin of early childhood environment to which the entire population of the United States is exposed, as well as early evidence on the mechanisms through which intergenerational transmission arises. However, given the nature of our central question, empirical approach, and data, it is impossible to entirely rule out the role of potential confounders in contributing to the observed estimates. Our analysis marks a first step towards understanding the intergenerational consequences of early-life environmental exposures that we hope will motivate new lines of inquiry.

³Assuming an upper bound intergenerational earning elasticity of 0.6, we calculate that the second-generation discounted lifetime earnings effect would capture less than 50% of the calculated lifetime earnings effect based on our estimated second-generation college attendance effect, and less than 25% of our estimated second-generation earnings effect.

⁴Black et al. (2019) evaluate the effects of in utero exposure to low-dose radiation on the later-life outcomes of the first generation as well as cognitive scores from military tests for the sons of the second generation in Norway. Nilsson (2017) focuses on the first-generation effects of alcohol availability on later-life outcomes in Sweden, but also explores health at birth for the second generation, finding no effect on health at birth. Heckman and Karapakula (2019) document the intergenerational effects of the Perry Preschool Program, a targeted anti-poverty program in the United States. Barr and Gibbs (2022) document the intergenerational effects of Head Start, another targeted anti-poverty program in the United States. East et al. (2023) document the effects of early-life Medicaid exposure on second-generation health at birth.

2 Background

What are the channels through which prenatal pollution exposure could have intergenerational effects? First, we need to understand how prenatal exposure to particulate matter could directly affect health and development. In our empirical analysis, we exploit variation in prenatal exposure to total suspended particles (TSP), the type of particulate matter that was regulated by the EPA at the time of the 1970s Clean Air Act Amendments (CAAA). TSPs represent a heterogeneous mixture of organic and inorganic compounds found in the air that are smaller than 100 microns in diameter. TSPs enter the atmosphere from both human activity (e.g., traffic, construction, and industrial production) and natural sources (e.g., wildfire smoke, sea salt, dust, and pollen).

All particulate matter can damage human health, although larger particles are believed to be less harmful than smaller particles (Dreher et al., 1996; Schwartz et al., 1996; Schwartz and Neas, 2000). Since larger particles settle more quickly than smaller particles, they are less likely to be inhaled. If larger particles are inhaled, they settle in the nose and throat and are usually expelled by coughing or sneezing. By contrast, smaller particles (i.e., those smaller than 10 microns) can remain in the air for extended periods of time and, once inhaled, penetrate the lungs and can cross the blood–brain barrier, causing inflammation in the central nervous system.

Once inside the body, particles are understood to affect respiratory function, lung development, and brain development (Anderson et al., 2011). Since particles can pass through the lungs into the bloodstream, they can also cause cardiovascular problems. Maternal exposure to particulate matter is understood to affect fetal health through a number of physiological pathways, such as oxidative stress, DNA damage to cells, damage to blood vessels, increased blood pressure, and reduced endothelial function, which could affect the transfer of nutrients to the fetus (Kannan et al., 2006; Erickson and Arbour, 2014). Particles can also be transferred directly to the fetus through the bloodstream, affecting respiratory, neurological, and cardiovascular development (Dejmek et al., 1999; Glinianaia et al., 2004; Bové et al., 2019;

Huifeng et al., 2020).

Particulate matter is understood to affect human capital and development throughout the life cycle. Its effects are not limited to the gestational period, although marginal damages are hypothesized to be greater during this period (Currie and Almond, 2011). This raises an important empirical challenge: we must isolate the effects of prenatal pollution exposure from contemporaneous effects experienced by the first and second generation throughout their life cycles. Simply comparing the outcomes of people who experience prenatal exposure to locations with cleaner air with those of people who experience prenatal exposure to locations with dirtier air would not identify the effects of prenatal exposure because individuals born in “treatment” locations may be exposed to lower air pollution over their entire life cycle, and so may their children. To distinguish between the effects of prenatal exposure and cumulative exposure within and across generations, we require a research design that compares individuals who have different exposure to air pollution during gestation but the same exposure postnatally. In Appendix B, we formalize this idea by extending the framework presented by Isen et al. (2017).⁵

The following section provides more details about the research design and empirical specifications that we use to estimate the relationship between parental prenatal pollution exposure and second-generation outcomes. We also provide direct evidence to support one of our key identification assumptions: that there is no differential effect of prenatal pollution exposure on later life pollution exposure.

3 Research Design

In this section, we provide an overview of the data and sample construction process, as well as the empirical specifications used to estimate the effects of parental prenatal particulate matter exposure on second-generation outcomes.

⁵The foundations of this model are based on earlier work by Grossman (1972), Bleakley (2010), and Cunha and Heckman (2007).

3.1 Data

To study the intergenerational effects of pollution exposure, it is necessary to identify the location and date of birth of each parent, infer their exposure to ambient air pollution, link these parents to their children, and measure outcomes. No single dataset has all of these features, and so our analysis requires linking Decennial Census, administrative records, and survey data. The data are linked using unique anonymous personal identifiers called Protected Identification Keys (PIKs), which can be thought of as “scrambled” Social Security numbers. PIKs are assigned to datasets using a probabilistic matching algorithm that links personally identifiable information (e.g., name, date of birth, Social Security Number, etc.) to a reference file of people in the United States.⁶

3.1.1 Parent–Child Links

We begin by assembling a database of parent–child links that are constructed using Decennial Census, administrative, and survey data sources available in the Census Bureau’s data linkage infrastructure. We identify links in two main datasets: the full-count Decennial Census from 2000 and 2010, and the American Community Survey (ACS) from 2005–2015. The set of links we are able to identify is not, we should stress, the full population of links. There are two main sets of parent–child linkages that we are not able to observe: parent–child links in households that formed and dissolved between Decennial Censuses (who were not ACS respondents) and parent–child links in which either the parent or child cannot be assigned a PIK.

We identify two types of parent–child link: “certain” links, where the survey and Census information allows us to identify a parent–child link with certainty, and “probable” links, where the survey and Census information allow us to identify the spouse of a certain link parent, who is also likely a parent (but may be a step-parent).

The Decennial Census and ACS data both contain detailed information on relationships

⁶For more on the process of PIK assignment, see [Wagner and Layne \(2014\)](#) and Appendix C.

within the household with one important limitation — the Decennial Census/ACS relationship question only collects information about the relationship between an individual and the head of household. This means that we can only identify certain parent–child links for the head of household. We additionally identify probable parent–child links between the head of household’s married or unmarried partner and the head of household’s children. For head of household–child links, we have additional information about the type of link — specifically, whether a child is biological, adopted, or a stepchild. In the main analysis, we focus on certain parent–child links to minimize any attenuation bias introduced through the incorrect assignment of children to affected or unaffected parents. In appendix materials, we show that our results are robust to incorporating probable links.

In total, we identify over 150 million parent–child links for parents born within 10 years of the CAA 1970. Only a fraction of these appear in our main analysis, which is based on a sample of the parent–child links that appear in the American Community Survey. We identify about 25 percent more “certain” links than probable and about 34 percent more mother links than father links. Appendix C describes the breakdown of our parent–child links in more detail.

3.1.2 Pollution Exposure

To analyze the intergenerational effects of pollution exposure, we need to infer the level of ambient air pollution to which the first generation was exposed during gestation. We do this in three steps. First, we link the set of unique parents described above with the Census Numident to obtain date and place of birth.⁷ We then obtain monitor-level daily pollution measures from the EPA, which we aggregate to the county level, and link these county-level measures to the parents’ place of birth.

Since place of birth data are not standardized, we assign county of birth information

⁷The Census Numident is a person-level administrative records file derived from the Social Security’s Numident, which contains all individuals who ever apply for a Social Security Number. The Numident contains information on the exact date and place of birth of individuals.

to first- and second-generation individuals using a crosswalk provided by Isen et al. (2017), combined with a probabilistic matching approach used in Voorheis (2017). For cases where county of birth cannot be assigned using the Isen et al. (2017) crosswalk, we match the Numident place of birth string to the United States Geological Survey’s list of places using a fuzzy string matching algorithm based on the optimal string alignment (OSA) distance metric.

We infer the level of pollution exposure experienced by individuals based on average pollution concentrations during the 270 days of gestation within their county of birth.⁸ To gather this pollution exposure information, we rely on EPA monitor data, which we retrieve using a public-facing API.⁹ Our pollutant of interest is particulate matter. For our period of study, the primary regulatory definition of particulates was total suspended particles (TSP). As discussed above, TSP represents a complex mixture of organic and inorganic compounds found in the air that are smaller than 100 microns in diameter, measured in units $\mu g/m^3$, which enter the atmosphere from both human activity (e.g., traffic, construction, and industrial production) and natural sources (e.g. wildfire smoke, dust, and pollen). We retrieve data from all TSP monitors between 1960 and 1980. During this time period, the monitoring network for criteria air pollutants was quite limited. This is a fundamental constraint to conducting research on the effects of air pollution during this time period. We use monitor-level observations to calculate the geometric mean and second-highest daily TSP concentration for all counties with at least one monitor in 1970. This results in data for 577 counties. Pollution concentrations in other counties were not measured during this time. These monitors were concentrated in urban areas, however, representing approximately two-thirds of the population in the contiguous United States in 1969 and covering all 50 states. TSP was one of the pollutants that was most widely monitored during this time. However, these data limitations result in a selected sample of predominantly urban populations and likely results

⁸It is possible that other moments of the pollution distribution may be important too. We focus on average gestational exposure to aid in comparison with the existing literature and mitigate measurement error concerns that may arise from missing data.

⁹See https://aqs.epa.gov/aqsweb/documents/data_mart_welcome.html for more details.

in measurement error in pollution exposure. Our instrumental variable approach helps to attenuate measurement error concerns.

For each county–day observation, we calculate the average TSP concentration across all active monitors. We then calculate county-level gestational period averages of prenatal TSP exposure for each birth.

3.1.3 Outcomes

Our main outcome measures are obtained from the 2005–2019 ACS, which contains detailed individual-level information on family structure, human capital, and labor market outcomes. The ACS is a nationally representative survey of a sample of households (about 1 percent per year), so we observe outcome measures for only a fraction of the parents and children identified above.

Our main outcome of interest is college attendance for the second generation. We measure college attendance contemporaneously — whether an individual was currently attending post-secondary education at the time of survey response. We focus on this outcome because very few second-generation individuals are old enough to evaluate earnings effects at the time of ACS response.¹⁰ By contrast, there is a much larger sample of second-generation individuals who are at the age of attending college and have parents that were born around the introduction of the 1970 Clean Air Act Amendments. We also explore high school completion as an alternative measure of educational attainment. Our main analysis sample reflects parent–child links where the child is aged 19–21 at the time of ACS response. We further restrict the sample to parents who were born between 1969 and 1975. Someone born in 1975 (the youngest cohort) could have been, in principle, 9–25 years old when someone in the second-generation sample was born.¹¹ For someone born in 1972, the age of birth would have been between 12–28. Although 60–70% of births occurred before the age of 29 in the

¹⁰Measures relating to “ever having attended college” or “completing college” are more relevant after the age of 25.

¹¹2019-19-1975 = 25. 2005-21-1975 = 9.

1990s and 2000s ([NCHS, 2022](#)), we caveat that our second-generation sample is more likely to represent the first-born children of younger parents.

There are several additional outcomes available to us that after preliminary consideration, we decided not to explore in the paper due to conceptual concerns. The ACS provides outcome measures related to disability. These disability measures are quite severe and do not have clear physiological pathways from air pollution. A priori, there is no reason to expect that early-life exposure to particulate matter should affect hearing, vision, ambulatory, self-care, or independent living. Another outcome that could, in principle, be explored is teen parenthood. However, we believe that the difficulties associated with codifying teen fertility outcomes in our data make this infeasible at present. In the ACS, parent-child links are identified using survey questions that measure relationships in reference to the main respondent. Since individuals who give birth as teenagers are much more likely to live with parents or other relatives while pregnant and after birth, we are only able to identify parent-child linkages for teen parents later in the life cycle, in cases where they become the first respondent in a different ACS survey and co-reside with their children. To evaluate teen fertility outcomes in a more systematic and comprehensive way, one would need access to the SSA's Kidlink database, which is currently not available to researchers at the U.S. Census Bureau ([Genadek et al., 2021](#)).

We construct later-life earnings outcomes for a different, much smaller sample of second-generation individuals who are 23 years of age and older. The number of observations in our analysis is almost half that of our main analysis sample. We caveat that the analysis using this sample is much more tentative. This sample is even more selected on the first-born children of relatively younger parents who were affected in the earliest years of the Clean Air Act. As with any analysis of "ever attending college" or "completing college", it is ultimately still too early in the life cycle of the second generation to fully evaluate the intergenerational earnings and labor market consequences of early-life pollution exposure.

In additional analyses, we explore several outcomes for the first generation with a view to

understanding the direct effects of prenatal pollution exposure and the mechanisms through which these effects could be transmitted across generations. Our main sample for the first generation analysis is ACS respondents who were born between 1969 and 1975, are aged 33–44 at the time of ACS response, for whom we are able to measure pollution exposure, and have at least one parent–child link. Economic outcomes are explored by defining variables for unemployment, public assistance receipts, and wages from detailed ACS questions on income and labor force participation. We also explore family structure, examining marriage, divorce, and fertility outcomes as we believe that these considerations could plausibly affect parental resources and investments. In evaluating family size, we use the full sample of ACS respondents during our sample period, not just those for which a parent–child link was formed. We do this to explore selection into the second-generation sample — whether pollution exposure affects the likelihood of a link existing, the number of links, whether a link exists in our second-generation college attendance sample specifically, and conditional on a link existing, the age at which the first link was formed. Finally, we look at how parents spend time with their children by linking the American Time-Use Survey with our existing data infrastructure. We caveat that the sample size is substantially smaller in this analysis because we are restricted to ATUS respondents born in monitor counties during our evaluation window. Following [Price \(2008\)](#), we construct a measure of “quality time”, which restricts the time spent with children to only include activities in which the child was the primary focus of the activity, or where a reasonable amount of interaction between parent and child would likely arise. This measure of quality time consists of 13 activities split into 4 categories: Category A - reading to/with, playing but not sports, helping with homework, helping/teaching, talking with/listening to, arts and crafts; Category B - eating meals together, Category C - playing sports, attending performing arts, participating in religious activities; Category D - looking after (as primary activity), physical care.

3.1.4 Controls

In addition to our outcome variables, the ACS provides sociodemographic information, including details on race, sex, and age. The first and second generation do not always appear in the ACS at the same time, so we also collect demographic characteristics from the Decennial Census to control for parental characteristics when evaluating second-generation outcomes. We also collect information on the characteristics of first- and second-generation county of birth — population, employment, personal income per capita, and total transfer income — from the Bureau of Economic Analysis’ Regional Product Accounts as control variables.

After linking these components together, we are left with two main analysis samples: 1) the second-generation sample of potential college attenders, where the unit of analysis is a parent–child link for all second-generation individuals aged 19–21 at the time of ACS response, and 2) the first-generation sample, which is at the individual parent level, linking place of birth and pollution exposure with outcomes for parents born 1969–1975 and aged 33–44 at the time of ACS response. All monetary values are expressed in 2008 dollars.¹²

3.2 Empirical Strategy

The objective of our main empirical exercise is to estimate the relationship between an individual’s prenatal particulate matter exposure and the later-life outcomes of their children 40–50 years later. There are two key threats to identification. First, exposure to particulate matter is correlated with many observable and unobservable characteristics that are correlated with long-run economic and social outcomes, resulting in omitted variable bias. We also face a second identification challenge, which is specific to identifying the effects of early-life shocks on later-life outcomes. We want to isolate the effects of prenatal exposure. Even if the shock was as good as randomly assigned, one would want to show that prenatal exposure to a shock does not affect exposure to future shocks later in life. If differences in early-life pollution exposure (or any early-life shock) are also correlated with differences in

¹²This is done to aid comparison with the earnings estimates in [Isen et al. \(2017\)](#).

later-life pollution exposure, it is not possible to disentangle the contribution of early-life shocks from later-life exposures. We must assume that the policy does not induce differential changes in later-life pollution exposure.¹³

To mitigate endogeneity concerns related to pollution exposure and maximize the likelihood of isolating the effects of early-life pollution exposure, we instrument for changes in particulate matter exposure using the introduction of the 1970 Clean Air Act Amendments (Chay and Greenstone, 2003a; Isen et al., 2017). The Clean Air Act was introduced in 1963 to regulate air pollution in the United States and is the largest environmental program in the country.¹⁴ It required the EPA to develop and enforce regulations to protect the population from exposure to airborne pollutants that are known to be hazardous to human health. In 1970, the Clean Air Act was amended, authorizing federal regulations to limit emissions, resulting in a major shift in the federal government’s role in air pollution control. As a consequence of the 1970 amendments, the EPA established the National Ambient Air Quality Standards (NAAQS), specifying the maximum acceptable levels for six criteria air pollutants — sulfur dioxide (SO₂), particulate matter (TSP, PM_{2.5}, and PM₁₀), nitrogen dioxide (NO₂), carbon monoxide (CO), ozone (O₃), and lead.¹⁵

The NAAQS standards were applied at the level of Air Quality Control Regions (AQCRs), which were collections of counties defined by the EPA. Any AQCR that exceeded these pollution levels was designated a “nonattainment area” and required to implement a plan to meet the standard. The EPA defined an AQCR as being in nonattainment of the NAAQS if any of the individual monitors in the AQCR fell outside the NAAQS standards. Counties with monitors out of attainment were required to improve air quality in line with the NAAQS, while counties that were in attainment did not, even if they were located in an AQCR that

¹³This assumption is required for any study that seeks to evaluate the later-life consequences of an early-life exposure or intervention.

¹⁴See Currie and Walker (2019) and Aldy et al. (2022) for a detailed history of the Clean Air Act.

¹⁵To the extent that our estimates capture correlated pollutants, a TSP interpretation may be too strong. In such a case, our results would be interpreted as capturing the effects of a compound “pollution exposure”; however, this is more of a concern for CO and ozone than for NO₂ and SO₂, which are the most common precursors to particulate matter.

was out of attainment (in other words, attainment counties in nonattainment AQCRs were inframarginal) — the NAAQS standards were binding at the county level. States worked with local governments and the EPA to develop state implementation plans (SIPs), which detailed the strategies used to achieve and maintain the NAAQs. Implemented strategies could vary substantially across localities within a state, depending on local sources of pollution (e.g., industrial facilities, power plants, construction, traffic). The consequences of not complying were severe. If a state did not act or developed an inadequate plan, then the EPA had the right to take control and implement their own plan. In this regard, the EPA acted as a “gorilla in the closet” (EPA, 1993), creating an incentive for states to self-enforce. Non-compliance could also lead to the withholding of federal funding for the state air pollution control program, highway construction, and sewage treatment plants. The EPA could also ban permits required for new or modified constructions that could source pollution. These powers are sufficiently broad that even the threat of regulatory action has been associated with reductions in pollution (Keohane et al., 2009).

We follow the existing literature and define treatment status based on the nonattainment status of each county, noting that it is the bindingness of EPA regulations and not the nonattainment designations themselves, which produce our quasi-experimental variation in pollution exposure.¹⁶ The EPA, however, did not publish county-level nonattainment designations until the late 1970s. Following Chay and Greenstone (2003a) and Isen et al. (2017), we reconstruct nonattainment designations using TSP concentrations from the years before the CAAA was enacted. Nonattainment of the primary air quality standard for TSP set in the 1970 CAAA occurs if the annual average (geometric mean) TSP concentration of a county is greater than $75 \mu\text{g}/\text{m}^3$, or the second-highest daily TSP concentration exceeds $260 \mu\text{g}/\text{m}^3$. As discussed above, we use monitor-level observations to calculate the geometric mean and second highest daily TSP concentration for all counties with at least one monitor

¹⁶While it is possible to define an AQCR-level treatment, almost all monitored AQCRs had at least one monitor out of attainment, meaning that an AQCR-level definition does not give meaningful variation in nonattainment designations.

in 1970. This allows us to categorize 258 counties as “nonattainment” counties and 319 counties as “attainment” counties.¹⁷

Following Chay and Greenstone (2003a) and Isen et al. (2017), we model the change in air pollution using an indicator variable for county nonattainment status interacted with an indicator for the years 1972 or later. The first-stage regression in this two-stage least squares estimator is a difference-in-differences regression model:

$$TSP_{j,c,t} = \alpha_0 + \alpha_1(Nonattainment_{c,1970} \times 1[\tau > 1971]) + \alpha_c^j + \alpha_{st}^j + \gamma X_j' + \delta X_c't + \nu_{j,c,t} \quad (1)$$

Exposure is measured for parent j , where $TSP_{j,c,t}$ is the average particulate matter concentration to which individual j was exposed during gestation in county c and year t , measured in $\mu g/m^3$. $TSP_{j,c,t}$ is regressed on a time-invariant county indicator equal to 1 if a county is designated as nonattainment, $Nonattainment_{c,1970}$, and interacted with an indicator equal to 1 for the years after the CAAA went into effect, $1[\tau > 1971]$. The interaction term is therefore equal to 1 for nonattainment counties following the implementation of the 1970 CAAA. The parameter of interest is α_1 , which provides a difference-in-differences estimate of the impact of nonattainment designation on prenatal exposure to TSP in the years after the CAAA regulations went into effect. Following Chay and Greenstone (2003a) and Isen et al. (2017), we also include additional controls. X_j' is a vector of individual characteristics, including age, race, and sex, as well as in utero weather conditions. $X_c't$ is a vector of county-level characteristics, measured in 1969, interacted with linear and quadratic time trends. α_c^j are county-of-birth fixed effects that control for time-invariant unobserved determinants of the labor market outcomes and family structure of parents born in county c . α_{st}^j are birth-state $\times$ birth-year fixed effects, which control for time-varying determinants of the long-run outcomes, common across all parents born in a state s in year t .¹⁸ Across all

¹⁷Our analysis is restricted to first-generation individuals born in these 577 counties and children of these individuals (the second generation).

¹⁸In extensions, we include second-generation controls, including second-generation birth-county fixed

specifications, we cluster our standard errors by the first generation’s county of birth — the level at which we measure exposure.

Consistent with previous research on the Clean Air Act, we show that the first-stage relationship is strong — that nonattainment status is associated with significant and persistent declines in particulate matter concentrations in the years after the 1970 CAAA came into effect. We estimate that, on average, nonattainment is associated with an $11.55 \mu\text{g}/\text{m}^3$ reduction in parental prenatal TSP exposure (Table 1, column 1). In panel a) of Figure 1, we present cohort-specific estimates of the effect of nonattainment on prenatal TSP exposure using a distributed-lag model. Following implementation, we estimate an immediate and persistent reduction in prenatal TSP exposure. Prior to the introduction of the 1970s CAAA, we estimate no statistically significant or economically meaningful differences between nonattainment and attainment counties, providing support for the parallel trends assumption. This inference is robust to extending the pre-treatment period back to 1967. Our main analysis restricts the pre-treatment period to start in 1969 to mitigate the influence of older pre-treatment cohorts who did not experience improvements in air quality until later. The influence of older cohorts leads to average differences that reflect broader improvements in air quality during the first years of life rather than isolating prenatal improvements.

We know, however, that there are limitations to evaluating the parallel trends assumption based on pre-treatment differences (Freyaldenhoven et al., 2019; Roth, 2022; Rambachan and Roth, 2023). We acknowledge these limitations by conducting additional sensitivity analyses following Rambachan and Roth (2023). Instead of imposing that the parallel trends assumption holds exactly, we bound how large post-treatment violations of parallel trends could be before inference “breaks down”. This is formalized by imposing that the post-treatment violation of parallel trends is no more than some constant, $\bar{M}$, larger than the maximum violation of parallel trends in the pre-treatment period. A value of $\bar{M}=1$, for example, imposes that the post-treatment violation of parallel trends can be no larger than the

effects, α_c^i , second-generation birth-state $\times$ birth-year fixed effects, α_{st}^i , and second-generation individual characteristics, X_t^i .

worst pre-treatment violation of parallel trends (accounting for statistical and identification uncertainty in our event-study estimates). For the estimated reduction in TSP, we estimate a breakdown value of $\bar{M}=0.7$ for the first post-treatment year, 1972. Consequently, our conclusion of a significant reduction in TSP depends on whether we are willing to restrict post-treatment violations of parallel trends to no more than 0.7 times as large as the maximal pre-treatment violation. Based on our pre-treatment estimates, this implies that our inferences are robust unless there are differential reductions in emissions from other shocks that can account for more than $2.25 \mu g/m^3$ (30% of the estimated differential reduction in 1972). Although we cannot rule out violations of parallel trends, these sensitivity tests make clear what must be assumed about possible violations of parallel trends to draw causal inferences.

In addition to providing evidence to support the parallel trends assumption, we show that there are limited differences between the characteristics of attainment and nonattainment counties prior to the 1970 CAAA (Table A1). In addition to several analyses that document balance and provide support for the research design, [Isen et al. \(2017\)](#) also show that a nonattainment designation is not correlated with any changes to the observable characteristics of mothers who gave birth in the years following the 1970 CAAA.

We also present evidence in support of the assumption that the research design identifies the effects of early-life pollution exposure. By restricting to a narrow window before the introduction of the 1970s CAAA, we can compare cohorts who have different TSP exposure during gestation but the same exposure post-birth. Columns 2-4 of Table 1 show that there is no meaningful effect of parental prenatal nonattainment exposure on first-generation later-life exposure at the time of ACS response (column 2), second-generation prenatal exposure (column 3), or second-generation later-life exposure (column 4). These findings are also supported in panels b), c), and d) of Figure 1. As with our first-stage estimate, we estimate no meaningful differences prior to the introduction of the 1970 CAAA. Unlike the effects on prenatal exposure, we estimate no meaningful differences in later-life TSP exposure between

nonattainment and attainment counties. This supports the premise of the research design — that children born in nonattainment counties before the 1970 CAAA came into effect still benefited from cleaner air later in life.¹⁹ We caveat that we are not able to rule out differences in post-natal childhood exposures for the first generation. We assign pre-natal exposures using location of birth; however, we do not have consistent information on residential location for individuals until the late 1990s. Because we cannot rule out early childhood effects, we refer to our treatment as more broadly reflecting early-life exposures.

In the second stage, we use the predicted TSP levels from Equation 1 in place of observed TSP levels,

$$College_{i,j,c,t} = \rho_0 + \rho_1 \widehat{TSP}_{j,c,t} + \alpha_c^j + \alpha_{st}^j + \gamma X_i' + \delta X_c' t + \varepsilon_{i,j,c,t,y} \quad (2)$$

Our specification is the same as Chay and Greenstone (2003a) and Isen et al. (2017), except that our outcomes are measured for child i born to parent j . The coefficient of interest, ρ_1 , captures the effect of a $10 \mu g/m^3$ increase in CAAA-driven first-generation prenatal TSP exposure on the later-life outcomes of the second-generation child i .

The main concern regarding the instrumental variables research design is the exclusion restriction. To identify the intergenerational effects of TSP exposure, it must be the case that the CAAA does not affect outcomes in any other way than through reductions in pollution. Isen et al. (2017) make the point that nonattainment designations could affect economic competitiveness (Henderson, 1996; Becker and Henderson, 1997; Greenstone, 2002; Greenstone et al., 2012; Walker, 2013). However, existing evidence suggests that the effects on the broader local economy are small, affecting less than 0.7 percent of the total workforce, and is largely based on the 1990 Clean Air Act Amendments (Walker, 2013). Since the 1970 CAAA was the first major regulation to be introduced, actions to reduce emissions may have been less costly in 1970 than in the 1990s, there being more low-hanging fruit, attenuating

¹⁹In Table A2 and Figure A1, we also show that there is little evidence of differential migration, an important mechanism through which later-life differences in pollution could arise.

concerns about economic competitiveness. Nevertheless, it is possible that the 1970 CAAA contributed to a decline in economic conditions for nonattainment counties, affecting the long-run economic prospects of affected individuals. Since effects on competitiveness would be expected to have opposite effects on health to reductions in pollution exposure, we argue that our 2SLS estimates will understate the intergenerational effects of pollution exposure if the exclusion restriction is violated. The reduced form effect of nonattainment remains valid and is interpreted as the intergenerational effects of the 1970 Clean Air Act Amendments instead of pollution. Our reduced form and corresponding 2SLS estimates produce conceptually identical results, providing some confidence that any residual violation of the exclusion restriction is unlikely to be a first-order concern.

We also consider the possibility of spillovers between nonattainment and attainment counties, that might exacerbate reductions in particulate matter (Becker and Henderson, 1997; Binish and Khanna, 2020). Even seemingly small spillovers, compounded over many years, can lead to significant effects in the long run. We argue, however, that any changes that manifest in persistent differences in economic activity should also manifest in terms of later-life differences in pollution exposure. As documented, we find no later-life differences in pollution exposure for the first generation or the second generation, and so it is unlikely that spillovers are a first-order concern in this context.

4 Results

Table 2 presents the results of our analysis on the relationship between first-generation early-life pollution exposure and the likelihood that the second generation attends college, using our preferred second-generation sample that uses certain parent–child links. We estimate that a $10 \mu\text{g}/\text{m}^3$ reduction in first-generation prenatal pollution exposure is associated with a 1.66–1.76 percentage point increase in the likelihood that the second generation attends

college.²⁰ This corresponds to a 3.8% increase in college attendance relative to the control mean. This is a meaningful effect, equivalent in magnitude to the second-generation effect of 0.098–0.121 Head Start Programs (Barr and Gibbs, 2022). In Panel B of Table 2, we present reduced form effects, capturing the effect of the 1970 CAAA on second-generation outcomes. We estimate that the reduced form effect of the 1970 CAAA is associated with a 1.92–2.08 percentage point increase in the likelihood of college attendance. Using the average nonattainment-induced changes in TSP for this sample ($11.55\mu g/m^3$), our IV estimates imply a 1.92–2.03 percentage point increase in the likelihood of attending college. The 2SLS and equivalent reduced form estimates are almost identical. Figure 2 presents cohort-specific estimates of the reduced form effect of nonattainment on second-generation college attendance. Following the introduction of the 1970 CAAA, we estimate an immediate and sustained 2 percentage point increase in college attendance for the 1972 cohort onwards; however, similar to Isen et al. (2017) the cohort-specific estimates are more noisily estimated than the pooled estimates. Similar to the pollution outcomes, we do not estimate any differential pre-treatment effects.

Our results are robust to including probabilistic links (Table A4). The estimates are marginally smaller in magnitude. This is consistent with the introduction of measurement error that would arise through the introduction of false links. The difference could also be attributable to an attenuated treatment effect when the parent–child link does not involve the reference person in the survey.²¹

We also conduct a permutation inference exercise to address any concerns that the results are driven by sampling variability or spurious trends. Such concerns may be reasonable in the context of exploring long-run or intergenerational effects, where even small amounts of confounding residual variation could be propagated over time if serially correlated, resulting

²⁰In Table A3, we estimate the effects for maternal exposure and paternal exposure separately. We estimate almost identical effects for paternal and maternal exposure and a 25% larger effect if both parents were exposed.

²¹Relationship information in the ACS is only asked in relation to the reference person; i.e., person 1 on the survey response.

in a spurious divergence of outcomes between treatment and control groups. Holding the sample fixed, we randomly reassign nonattainment status across counties 1,000 times and use these placebo realizations of nonattainment to estimate the original model. We then compute the share of placebo estimates that are higher in absolute value than our original estimate and calculate an exact p-value. For the association between first-generation prenatal nonattainment exposure and first-generation prenatal TSP exposure (our first stage), we calculate an exact p-value of 0.000. For the association between first-generation prenatal nonattainment exposure and second-generation college attendance, we calculate an exact p-value of 0.0136. We calculate smaller exact p-values when randomly reassigning the year that nonattainment came into effect, holding nonattainment status fixed, or randomizing both dimensions of treatment. These findings provide compelling evidence that our estimates are not driven by sampling variability or spurious trends.

We caveat again that our estimates are based on what could be a selected sample. The children in our second-generation analysis sample are aged 19-21 and so are more likely to be the first-born children of relatively younger parents. This may affect the degree to which our results generalize to all second-generation individuals. However, it is not possible to sign the selection bias. If first-born children are more likely to attend college, the marginal effect for this sample would be smaller than the full population. If children born to younger parents are less likely to attend college, the marginal effect for this sample may be larger than the full population.

In addition to college attendance, we look at several additional outcomes in Table A5 and Figure A2. In column 1, we report the relationship between first-generation prenatal pollution exposure and the likelihood of second-generation high school completion. We restrict the sample to be the same as our main analysis sample. Unlike college attendance, we do not estimate any meaningful effect on the likelihood of high school completion. We note that completing high school is a high probability outcome (84% of our control group completed high school) and so it may take a more substantial treatment to see a response

on this margin compared to college attendance (where only 44% of the control group were attending college at the time of ACS response). In columns 2–4, we provide early evidence on the intergenerational earnings effects of parental prenatal pollution exposure. To do this we have to rely on a different, much smaller, sample of parent–child links, where the second-generation individuals are 23 years old and above. The number of observations in our analysis is almost half the size of our main analysis sample. We estimate meaningful but noisy estimates of first-generation prenatal TSP exposure on second-generation earnings. Taking the coefficients at face value, a $10 \mu\text{g}/\text{m}^3$ reduction in parental prenatal pollution exposure is associated with a 1%, or \$392, increase in later-life earnings for the second generation (columns 2 and 3). If we include second-generation individuals with zero earnings in the sample, the magnitude is larger, equivalent to \$641, and is statistically significant at the 10% level in the reduced form analysis. These findings provide initial, suggestive evidence that early-life pollution exposure has effects on the later-life earnings of the second generation; however, ultimately, it is still too early in the life cycle of the second generation to fully evaluate intergenerational earnings and broader labor market effects.

Taking these numbers at face value, we consider what they imply for lifetime earnings. The following numbers reflect simple back-of-the-envelope calculations and should not be interpreted causally. Assuming a 3 percent real discount rate (5 percent discount rate + 2 percent wage growth), we calculate an $\$1,101/10\mu\text{g}/\text{m}^3$ increase in second-generation cumulative lifetime earnings, discounted to age zero of the first generation, based on our estimated coefficient in column 3 of Table A5.²² By contrast, if we were to assume a college wage premium of \$28,547 p.a. (estimated using the public use CPS-ASEC), then combined with a graduation rate of 50 percent to convert college attendance to college completion²³, and a 3 percent real discount rate we calculate a $\$666/10\mu\text{g}/\text{m}^3$ increase in second-generation cumulative lifetime earnings, discounted to age zero of the first generation. Combining the number of children born in nonattainment counties in 1972 (1.52 million), the probability that one

²²We choose a 3 percent real discount rate to be consistent, and aid comparison, with Isen et al. (2017).

²³See https://nces.ed.gov/programs/coe/indicator_ctr.asp

of these individuals had a child (0.63), and the average number of children that were born to these individuals (2.24), we calculate that the total second-generation earnings effects for the 1972 cohort are \$1.43–2.36 billion dollars. If reductions in TSP were persistent (and evidence suggests they were), we would expect these benefits to be realized for each cohort since 1972. The smaller number captures intergenerational effects on income that come only from second-generation college attendance. The larger number captures second-generation college attendance and any other second-generation human capital benefits that aren’t capitalized into the “college-wage premium”. However, we reiterate our caveat that the direct estimates on earnings are noisily estimated and based on a sample of young workers. It is unclear whether they reflect an upper or lower bound on second-generation earnings. We leave a more complete analysis of intergenerational earnings to future research.

4.1 Exploring Mechanisms

We posit that there are two broad channels through which the consequences of prenatal pollution exposure could propagate across generations. The first channel is biological. Gestational pollution exposure could result in epigenetic changes, i.e., permanent changes in gene expression.²⁴ If so, hereditary transmission of these changes could affect the second generation. It is also possible that pollution exposure could affect the reproductive health and birth outcomes of women without any genetic or epigenetic changes, resulting in subsequent health effects for the second generation. The second mechanism is economic. If early-life pollution exposure affects the human capital and earnings of the first generation, then this could plausibly affect parental resources and investments that contribute to the production of second-generation human capital.

²⁴Epigenetics is the study of heritability not related to DNA sequencing. There is a strong theoretical basis for the conjecture that environmental shocks could lead to permanent changes in gene expression (Jablonka and Raz, 2009; Lim and Brunet, 2013; Heard and Martienssen, 2014). In addition, evidence from animal studies shows that environmental shocks during pregnancy affect not only the immediate offspring, but later generations as well, even if immediate offspring receive remedial care (Francis et al., 1999; Dunn and Bale, 2009; Jimenez-Chillaron et al., 2009; Rehan et al., 2012; Radford et al., 2014).

4.1.1 Biological Effects

To explore the empirical relevance of the hereditary mechanism, we examine the differential effect of first-generation early-life pollution exposure between biological and non-biological (adopted and step) second-generation children. This exercise provides an opportunity to examine the extent to which permanent changes in gene expression, triggered by prenatal pollution exposure, are passed down from parent to child. These changes in gene expression could affect human capital, and in turn the likelihood of, and returns to, attending college. As the Decennial Census and ACS both ask whether the child of the main respondent is biological, adopted, or step-child, we are able to identify a set of parent–child links for which there should be no direct hereditary mechanism.

Biological children could be affected through both changes in the household environment and through genetic channels. By contrast, adopted and step-children should only be affected through the household environment (i.e., parental resources and investments). If the effects on college attendance are entirely driven by genetic pathways, we would expect there to be no effect on adopted children. The differential effect would be the inverse of the effect on biological children. If the college attendance effect is entirely driven by the household environment, there should be no differential effect on adopted children. We note that it is entirely possible that our estimated effects partially reflect both channels, at which point the effect on adopted children should be smaller than the effect on biological children.

We have considered three potential concerns with this exercise. First, there may be a differential propensity to have non-biological children in nonattainment counties, resulting in selection into the second-generation sample. For example, if prenatal exposure to pollution reduces fertility or fecundity, the children of parents who benefited from reductions in prenatal pollution may be more likely to be biological. In column 1 of Table 3, we show that the relationship is small and statistically insignificant. Taken at face value, we estimate that a $10\mu\text{g}/\text{m}^3$ reduction in first-generation prenatal TSP exposure is associated with a 0.14 percentage point increase in the likelihood that a parent–child link in our college attendance

sample is not biological — a 0.7% increase compared to the control mean. A second concern is that adopted children may be affected through hereditary channels if their biological parents were born in nonattainment counties at the same time as their adoptive parents. However, this requires that their adoptive parents and biological parents are born at the same time and location, which we argue is unlikely to be the case in a systematic way. In support of this assumption, we observe that adopted children are more likely to be born in a different county from their adoptive parents than biological children. Finally, implicit in this analysis is that non-biological and biological children/parents are the same in other dimensions. It is possible that there could be differential investments due to differences in the characteristics of parents or children. To mitigate this, we compare treated biological parent–child links to control biological parent–child links and do the same for non-biological parent–child links; however, we cannot rule out that differences in marginal damages could reflect differential investments, undermining a hereditary interpretation.

Columns 2 and 3 of Table 3 estimate the relationship between parental prenatal TSP exposure and second-generation college attendance for biological and non-biological children. We estimate that lower first-generation prenatal pollution exposure is associated with a meaningful increase in the likelihood of attending college for both biological and non-biological children, although the estimate is only statistically significant for biological parent–child links. The sample size of non-biological parent–child links is smaller, which may contribute to the difference in statistical significance; however, the magnitude is also smaller. For biological parent–child links, a $10\mu g/m^3$ reduction in first-generation prenatal TSP exposure is associated with a 1.91 percentage point increase in the likelihood of attending college — a 4.3 percent increase relative to the control mean. For non-biological parent–child links, a $10\mu g/m^3$ reduction in first-generation prenatal TSP exposure is associated with a 1.35 percentage point increase in the likelihood of attending college — a 3.4 percent increase relative to the control mean. While confidence intervals overlap, our central

estimates suggest that the heritable channel could account for 30% of the overall effect.²⁵

These findings provide suggestive evidence that the estimated college attendance effect is more likely to be driven by parental resources and investment channels, than hereditary channels (Bjorklund and Chadwick, 2003; Bjorklund, 2006; Bjorklund et al., 2007, 2010; Black et al., 2020; Kleven et al., 2021a). In the following section, we explore the empirical relevance of different non-hereditary mechanisms.

4.1.2 Income Effects

Turning to economic channels, we explore the relationship between early-life pollution exposure and first-generation labor market outcomes. Table 4 presents the results of this analysis.²⁶

In column 1, we estimate that a $10 \mu\text{g}/\text{m}^3$ reduction in prenatal TSP exposure is associated with a statistically insignificant 0.8 percent increase in earnings at the time of ACS response. Evaluated at the mean earnings, this corresponds to an annual earnings effect of \$344 compared to the control mean. A similar inference is drawn when estimated in levels. The magnitudes of our estimates are slightly smaller than the estimates presented in Isen et al. (2017). Differences in these effects could be driven by differences in the age of evaluation, as well as the quality of the earnings data. Isen et al. (2017) estimate effects at age 29-31 using administrative earnings data from the 24 Longitudinal Employer-Household Dynamics program (LEHD) states. Our sample contains parents in all states and evaluates self-reported earnings at age 33-44.²⁷

In Table B3, we estimate a larger relationship between early-life pollution exposure and

²⁵ $1 - (1.35/1.91) = 0.294$.

²⁶Cohort-specific estimates are presented in Figure B1. We do not estimate any meaningful differential trends prior to the introduction of the 1970 CAAA. This provides further support for the parallel trends assumption.

²⁷If we take the earnings effect at face value, assume it is constant over the life cycle, and assume a real discount rate of 3 percent back to age zero, we calculate that the lifetime earnings effect of a $10 \mu\text{g}/\text{m}^3$ increase in TSP is \$1,768 per person (\$2.69 billion for the 1972 cohort). If we were to use the central estimate from Isen et al. (2017), we would calculate a lifetime earnings effect of \$2,541 per person (\$3.86 billion for the 1972 cohort).

later-life earnings for first-generation fathers that are in line with the magnitudes presented in Isen et al. (2017), however, they remain statistically insignificant at conventional levels. By contrast, we estimate a much smaller earnings relationship for first-generation mothers (Table B2). Isen et al. (2017) do not estimate differential earnings effects by sex, suggesting that part of the difference in our findings could be driven by restricting our sample to parents and evaluating earnings later in the life cycle, after more workers have had children. If treated mothers were more likely to work than control mothers, as suggested by the increased number of quarters worked in Isen et al. (2017), then the child penalty could contribute to a convergence of earnings effects for treated vs. control mothers when looking at earnings at ages 33–44 after women have had children (Kleven et al., 2021b).

In addition to looking at earnings, we also explore the relationship between early-life pollution exposure and labor force participation (column 3), unemployment (column 4), and receipt of public assistance (column 5). The sample size for these outcomes is slightly larger, as they include individuals with zero earnings. We fail to reject the null hypothesis that there is no relationship between gestational TSP exposure and these outcomes. The magnitude of the coefficients are very small. The absence of any effects suggest that TSP exposure does not appear to affect labor market outcomes on the extensive margin. In Appendix B we explore first-generation educational outcomes. We fail to reject the null hypothesis that prenatal TSP exposure has no effect on years of schooling or college attendance (Table B1 and Figure B2). This is consistent with Isen et al. (2017), who argue that the estimated effect on earnings is too small to be driven by schooling.²⁸ Overall, we find little evidence that early-life pollution exposure has much of an effect on the measurable labor market outcomes of the first-generation.

While we do not estimate a statistically significant relationship between early-life pollu-

²⁸In Tables B4 and B5, we do not estimate any differential effect on first-generation educational attainment by sex. Educational differences do not explain the differential relationship between early-life pollution exposure and later-life earnings by sex, consistent with the premise that the flattening of the earnings response for women relates to gender-specific changes in earnings patterns after the period of analysis covered by Isen et al. (2017).

tion exposure and first-generation later-life earnings, the similarity in magnitudes with [Isen et al. \(2017\)](#) suggest that greater resource availability when children were younger could contribute to our estimated second-generation college attendance effects. However, the existing evidence on the relationship between parental wealth and college attendance suggest that our college attendance effect is unlikely to be driven solely by parental resources. [Lovenheim \(2011\)](#) and [Lovenheim and Lockwood Reynolds \(2013\)](#) explore the effects of an increase in household wealth on college attendance, finding that a \$10,000 increase in household wealth is associated with a 0.71-0.92 percentage point increase in the likelihood of attending college. [Bulman et al. \(2021\)](#) explore the effects of winning the lottery on college attendance. They estimate a 0.6 percentage point increase in attending college per \$100,000 increase in household resources. If we take the largest estimate of the effect of household wealth on college attendance from the existing literature, assume that both parents were treated by the reductions in air pollution, and assume that our findings were driven entirely by the increase in household earnings, then we would expect a 0.33 percentage point increase in the likelihood of attending college (32% of our most conservative estimate).²⁹

To fully reconcile our results using central estimates, we would need it to be the case that household income in early childhood has a much larger effect on college attendance – more than 5 times larger than [Lovenheim and Lockwood Reynolds \(2013\)](#) (80 times larger than [Bulman et al. \(2021\)](#)). However, there are differences in our context, compared to the existing literature, that may provide support for income as a more important driver. In the existing literature potential college attendees are usually in their teenage years at the time that the wealth shock is realized. For example, [Lovenheim \(2011\)](#) and [Lovenheim and Lockwood Reynolds \(2013\)](#) look at wealth shocks in the 4 years prior to turning 18, and most of the first-born children in [Bulman et al. \(2021\)](#) are teenagers at the time that their parents win the lottery. It is entirely possible that wealth shocks during the teenage years have much less of an effect on college attendance since many important human capital

²⁹ $\$1,768 \times 2 = \$3,536 = 35.36\%$ of \$10,000. $0.3536 \times 0.92 = 0.325$.

decisions have already been realized (Keane and Wolpin, 1997; Cunha et al., 2006; Currie and Almond, 2011; Almond et al., 2018). Drawing on insights from the early childhood literature, it is plausible that a parental wealth shock in early-life could have a larger effect on college attendance due to dynamic complementarities. For example, Caucett and Lochner (2020) show that income or wealth shocks that occur earlier in parents’ lives, or that are at least anticipated when children are younger, can have much larger effects on human capital investments than surprise income shocks when children are older. If we take the upper bound of the 95% confidence interval from Lovenheim and Lockwood Reynolds (2013) — a 1.5 percentage point increase — we would calculate a 0.53 percentage point increase in the likelihood of attending college (32% of our most conservative estimate).³⁰

Our findings, combined with the best available evidence to date, suggest that the total benefits of early-life pollution reductions may not be fully capitalized into first-generation earnings. Absent a direct evaluation of the second-generation effects, we would substantially underestimate second-generation college attendance and the implied second-generation earnings effect. A naive calculation of second-generation college attendance and earnings effects using first-generation estimates and central (0.34) and upper bound (0.6) estimates of the intergenerational earnings elasticity from Chetty et al. (2014) delivers a second-generation discounted lifetime earnings effect between \$180 and \$313 per person (\$0.39-0.67 billion for the 1972 cohort). These calculations captures less than 27–47% of the calculated earnings effect based on our estimated second-generation college attendance effects, and less than 17–28% of our estimated second-generation earnings effects. These insights highlight the importance of directly evaluating intergenerational effects and suggest that other mechanisms, such as parenting-specific human capital, may be important in explaining the intergenerational effects of early-life pollution exposure.

³⁰\$1,768 = \$3,536=35.36% of \$10,000. $0.3536 \times 1.5 = 0.53$.

4.1.3 Family Structure and Fertility Effects

Another possibility is that part of our estimated college attendance effect reflects selection into the second-generation sample through changes in fertility. If those that were exposed to lower levels of pollution are richer and healthier, this may affect the likelihood of family formation through marriage and/or fertility choices. [Becker \(1960\)](#) posits that wealthier parents may choose to invest more in the children they have instead of having more children. However, empirical evidence suggests that increases in earnings or wealth are associated with increases in fertility ([Lovenheim and Mumford, 2013](#); [Black et al., 2013](#); [Dettling and Kearney, 2014](#); [Kearney and Wilson, 2018](#); [Autor et al., 2019](#)). If, despite the existing literature, parents that were exposed to lower levels of TSP have smaller families, then under the plausible assumption that family size is negatively correlated with the likelihood of college attendance, this could explain our results.

Health and income differences may affect the likelihood of family formation, affecting the likelihood that the first generation has children, or the household environment in which children are born. It is also possible that pollution exposure could affect fertility or fecundity ([Carre et al., 2017](#); [Clay et al., 2021](#)). In such a case, there may be differential selection into the second-generation sample between the treatment and control groups.

Table 5 presents our findings on the relationship between early-life pollution exposure and fertility and family structure.³¹ We fail to reject the null hypothesis that there is no difference in the likelihood of a parent–child link existing (column 1), or in the number of parent–child links associated with a first-generation individual (column 2). In column 3, we directly explore the extent to which there is differential selection into our main analysis sample. Given their age, these second-generation individuals are more likely to be the first child of relatively younger parents. If treated first-generation individuals are differentially likely to form relationships and have children earlier in the life cycle, then this could differentially

³¹Cohort-specific estimates are presented in Figure B3. Again, we do not estimate any meaningful differential trends prior to the introduction of the 1970 CAAA, providing further support for the parallel trends assumption.

affect the likelihood of college attendance in our main analysis. We do not estimate any differential likelihood of a parent–child link being found in this sample between the treatment and control group. We also don’t estimate any difference in the age that the first parent–child link was formed between treatment and control individuals (column 4), or the likelihood of marriage (column 5). Taken at face value, the magnitude of all estimates are small in absolute terms and relative to the control mean. This suggests that selection into the existence of second-generation individuals, both overall and in our main analysis sample, is unlikely to be a first-order concern.

In column (6) of Table 5, we estimate that a reduction in prenatal TSP exposure is associated with a reduction in the likelihood of divorce. This may point to an increase in the quality of marriages, resulting in a more secure family environment. The magnitude of the effect is small, corresponding to a 0.36 percentage point decrease in the likelihood of being divorced per $10 \mu g/m^3$ reduction in prenatal particulate matter exposure. While previous research suggests that the children of divorce may be substantially less likely to attend college (Painter and Levine, 2000; Gruber, 2004; Frimmel et al., 2024), the magnitude of the effect suggests that divorce could possibly explain up to 2% of our most conservative college attendance result.³²

One might be concerned that the average effects presented in Table 5 are a net zero if income effects result in differential fertility responses for men and women. Economic theory suggests that improvements in male labor market conditions should be associated with increases in fertility, while better wages and employment opportunities for women should have opposing income and substitution effects (Schaller, 2016). In Tables B6 and B7, we explore whether there are differential family structure effects by sex. We fail to reject the null hypothesis that early-life pollution reductions have no effect on the creation of parent–child links, the number of parent–child links, the age at which a parent first has a child, or the

³²We assume that a 1 percentage point increase in the likelihood of divorce is associated with a 0.1 percentage point decrease in the likelihood of attending college. We caveat that causal evidence on the effect of divorce is limited, reducing our confidence in this calculation.

likelihood of marriage. We do estimate a small differential effect in the likelihood that women have a parent–child link in the second–generation sample; however, given the identical effect of maternal or paternal exposure on the likelihood of second-generation college attendance, there is no gender-specific selection to explain (Table A3). We also see that the estimated divorce effect is concentrated among men, indicating that first-generation treated fathers are more likely to be in the household.

Finally, in Table B8, we explore whether reductions in particulate matter exposure affected who individuals married, an examination of assortative matching. We estimate that a $10 \mu\text{g}/\text{m}^3$ reduction in prenatal TSP exposure is associated with a 12 percentage point increase in the likelihood that an individuals’ partner was treated, but little in the way of meaningful relationships between early-life pollution exposure and other partner characteristics.

Collectively, our findings suggest that fertility-driven differential selection into the second-generation is unlikely to explain our main results.³³ The estimated relationship between early-life pollution exposure and divorce suggests a role for family structure in shaping second-generation outcomes but is arguably too small in magnitude to drive the results.

4.1.4 Parental Investments

Traditionally, models exploring the mechanisms underlying intergenerational persistence have implicitly assumed that all parents are equally capable. More recently, [Cunha and Heckman \(2007\)](#), [Heckman and Mosso \(2014\)](#), and [Becker et al. \(2018\)](#) model increases in parental human capital as having an effect not only on the earnings of parents, but also on the production of their children’s human capital, introducing an additional mechanism

³³One may still be concerned about selection into the first-generation sample given the results by [Chay and Greenstone \(2003a\)](#), documenting that the 1970 CAAA-induced changes in TSP reduced infant mortality rates. This could contribute to differential first-generation outcomes, and consequently, second-generation outcomes due to avoided mortality effects for marginal first-generational individuals. [Isen et al. \(2017\)](#) estimate that mortality-induced selection is very small, likely increasing the size of the labor force by 0.16 percent. In bounding exercises, they do not estimate any meaningful differences to their estimated effects on labor market outcomes. As such, we think it is unlikely that selection into the first-generation sample contributes meaningfully to our results.

— parental investments. Within economics, the idea that parental human capital might be important dates back to [Marshall \(1890\)](#) who wrote “The most valuable of all capital is that invested in human beings; and of that capital the most precious part is the result of the care and influence of the mother.” If reductions in early-life pollution exposure increased the first generation’s health as well as wealth, then as parents, they may be better placed to spend time and make investments in their children, improving outcomes for the second generation ([Murnane et al., 2000](#); [Heckman and Carneiro, 2003](#); [Cunha et al., 2010](#); [Del Boca et al., 2013](#); [Heckman et al., 2013](#); [Lundberg, 2017](#)).

One concrete approach to understanding the role that parental investments in cognitive and non-cognitive skills might play is to examine the relationship between early-life pollution exposure and parental time-use. We caveat that the sample is considerably smaller due to the small sample size of the American Time Use Survey as well as the need to restrict the sample to respondents born around the introduction of the 1970 CAAA. We are left with a sample size of around 5,600 first-generation observations. We further caveat that these parent-child experiences do not necessarily reflect the parent-child experiences of the children in our second-generation college attendance analysis due to differences in age and timing.

Table 6 presents estimates of the relationship between early-life pollution exposure on the total amount of quality time spent with children (column 1) and on each of the four categories defined by [Price \(2008\)](#) (columns 2-5).³⁴ We estimate that parents that experienced reductions in early-life pollution exposure spend more quality time with their children. The reduced form estimates and instrumental variable estimates are similar in magnitude, but the reduced form estimates are more precisely estimated. A $10 \mu g/m^3$ decrease in prenatal pollution exposure is associated with an additional 25 minutes of parent-child quality time each day. This is a 20% increase relative to the control mean. This increase is largely driven

³⁴Figure B5 presents cohort-specific estimates. Consistent with our other results, we do not estimate any meaningful differential trends prior to the introduction of the 1970 CAAA, providing further support for the parallel trends assumption. The overall effect on quality time (panel A) is more noisily estimated. The categories responsible for driving the overall effect in Table 6 are more precisely estimated.

by time spent eating meals together and time spent engaging with the arts. We reiterate that these results do not speak directly to the experiences of the second-generation individuals in our college attendance analysis, due to differences in age and timing. However, the identification of a parent–child quality time relationship for younger second-generation individuals helps to mitigate concerns that our second-generation results might not apply to children born later or to older parents.

To the best of our knowledge, we are not aware of any studies that evaluate the contribution of parent–child quality time on college attendance. However, several studies have shown that parent–child quality time is associated with increases in cognitive and non-cognitive skills and high-stakes test scores required for college attendance. These findings provide suggestive evidence that parental time could be a potentially important contributor (Neidell, 2000; Cunha et al., 2006; Todd and Wolpin, 2007; Del Bono et al., 2016; Gould et al., 2020). However, given the limitations and suggestive nature of this analysis, more research is required to understand the later-life consequences of parent–child quality time.

5 Conclusion

We provide early, quasi-experimental evidence on the intergenerational consequences of early-life exposure to ambient air pollution. Exploiting variation in particulate matter induced by the introduction of the 1970 Clean Air Act Amendments — which substantially reduced ambient air pollution — we estimate that the children of affected cohorts born in nonattainment counties after its implementation were more likely to attend college 40-50 years later than those born before. Supplementary analyses provide suggestive evidence that heritability could account for up to 30% of our estimated second-generation effect and that additional financial resources in earlier childhood could, at most, contribute up to 32%. Further research to understand how additional resources at different childhood stages affect later-life outcomes is needed. We also provide suggestive evidence that differences in parenting practices may

contribute, although these findings are based on a much smaller sample of second-generation individuals born after the second-generation individuals that are represented in our main sample.

Back-of-the-envelope calculations suggest that the total first- and second-generation cumulative lifetime earnings effects are 50-85% larger than the total first-generation cumulative lifetime earnings effects for the 1972 cohort. This combined value is equivalent in magnitude to 40–60% of the mortality benefits estimated in [Chay and Greenstone \(2003a\)](#). These values are substantially larger than naive calculations based on a combination of first-generation earnings effects and existing intergenerational elasticities. This highlights the importance of directly evaluating intergenerational effects. To date, the monetized mortality benefits of environmental regulations constitute the vast majority of overall benefits in benefit–cost analysis. Our findings show that the returns to investments in environmental quality are much larger than previously understood.

It is striking that these effects arise from such short exposure times early in life. If relatively small changes in air pollution have intergenerational consequences through parental resources and investments, it is likely that larger shocks and policies also have intergenerational consequences. This reinforces the potential benefits that could be realized through a reallocation of resources from later to earlier in the life cycle. Understanding the mechanisms through which such reallocations could be delivered remains an important area for future research. Our newly constructed data infrastructure presents exciting opportunities to engage more systematically with such questions.

Our results also have potentially important implications for inequality and economic opportunity. It is well established that individuals who live in low-income areas and disadvantaged communities are disproportionately exposed to pollution and other environmental risks ([Commission for Racial Justice, United Church of Christ, 1987](#); [Mohai et al., 2009](#); [Banzhaf et al., 2019](#); [Colmer et al., 2020, 2021](#); [Currie et al., 2023](#)). We document that early-life exposure to pollution has intergenerational consequences that are likely propagated through

economic resources and investments. Our results open up new lines of inquiry, raising questions about the role that investments in environmental quality might play in shaping upward mobility, economic opportunity, inequality, and the intergenerational persistence of poverty.

We note caveats. First, many of the individuals in our second-generation analysis sample are first-born children, born to younger parents. While we do not find any evidence that selection into the second-generation sample is related to first-generation pollution exposure, our results may not necessarily generalize to second-born children or to children born to older parents. Second, we do not provide a full understanding of the intergenerational effects of air pollution. We provide early evidence that first-generation early-life pollution exposure is associated with one margin of second-generation human capital accumulation. Given these caveats, we conclude that the estimated effects on college attendance are strongly suggestive of important intergenerational transmission effects. This conclusion is supported by early evidence that reductions in first-generation pollution exposure may translate into higher earnings and employment in the second-generation. More research is needed to conclusively understand the intergenerational consequences of early-life pollution exposure. We hope that the new data linkages that we have constructed will facilitate additional research in this area.

Data Availability

Code replicating the tables and figures, as well as information about the proprietary data used in this article is provided by [Colmer and Voorheis \(2025\)](#) and can be found in the Harvard Dataverse, <https://doi.org/10.7910/DVN/PVNEZV>.

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Tables and Figures

Table 1: The Relationship between Nonattainment and Pollution Exposure Over the Life Cycle

	(1) FIRST-GEN PRENATAL EXPOSURE ($\mu g/m^3$)	(2) FIRST-GEN LATER-LIFE EXPOSURE ($\mu g/m^3$)	(3) SECOND-GEN PRENATAL EXPOSURE ($\mu g/m^3$)	(4) SECOND-GEN LATER-LIFE EXPOSURE ($\mu g/m^3$)
Nonattainment $\times$ Post	-11.55*** (3.095)	-0.175 (0.416)	0.425 (0.256)	0.300 (0.412)
First-Gen Fixed Effects	YES	YES	YES	YES
First-Gen Prenatal Controls	YES	YES	YES	YES
Observations	139,000	46,000	106,000	43,000
Control Mean	91.00	46.35	54.50	41.89

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a parent-child link for the sample of second-generation individuals that were aged 19–21 at the time of ACS response. The outcome of interest in column (1) is first-generation prenatal TSP exposure. The outcome of interest in column (2) is first-generation later-life TSP exposure, measured at ACS response. The outcome of interest in column (3) is second-generation prenatal TSP exposure. The outcome of interest in column (4) is second-generation later-life TSP exposure, measured at age 16. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year), first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the first-generation individual's nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclos practices. Standard errors are clustered at the first generation's county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table 2: The Relationship between First-Generation Prenatal TSP Exposure and Second-Generation College Attendance

	SECOND-GENERATION COLLEGE ATTENDANCE		
	(1)	(2)	(3)
Panel A: IV			
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0176** (0.0073)	-0.0166** (0.0083)	-0.0173** (0.0083)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.0208*** (0.0072)	0.0192** (0.0078)	0.0198** (0.0077)
First-Gen Fixed Effects	YES	YES	YES
First-Gen Controls	NO	YES	YES
Second-Gen Controls	NO	NO	YES
Second-Gen Fixed Effects	NO	NO	YES
Observations	139,000	139,000	139,000
Control Mean	0.438	0.438	0.438
First-Stage KP F-Stat	20.17	13.93	14.66

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a parent-child link. The analysis sample is parent-child links where the second-generation individual was aged 19–21 at the time of ACS response. The outcome of interest in all columns is an indicator for whether the second-generation individual is attending college at the time of their response to the American Community Survey (2005-2019). Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. Column (1) reports estimates from the most parsimonious specification, which includes first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No additional controls are included. Column (2) reports estimates that control for first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the first-generation individual's nine month gestational period. Column (3) further includes second-generation fixed effects (county-of-birth, state-of-birth $\times$ year, and birth month) and second-generation demographic controls (indicators for sex, race/ethnicity, and age). The number of observations are rounded in concordance with US Census Bureau disclosure practices. The underlying sample size is constant across columns. Standard errors are clustered at the first generation's county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table 3: The Relationship between First-Generation Prenatal TSP Exposure and Second-Generation College Attendance: Adopted and Step-Children vs. Biological Children

	ADOPTED OR STEP-CHILD (1)	SECOND-GENERATION COLLEGE ATTENDANCE (2)	(3)
Panel A: IV			
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0014 (0.0048)	-0.0191** (0.0093)	-0.0135 (0.0143)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.00157 (0.0056)	0.02188** (0.0086)	0.01515 (0.0152)
Relationship	—	Biological	Adopted/Step-Child
First-Gen Fixed Effects	YES	YES	YES
First-Gen Controls	YES	YES	YES
Observations	139,000	113,000	26,000
Control Mean	0.186	0.448	0.395
First Stage KP F-Stat	13.93	13.34	14.35

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a parent-child link. The analysis sample is parent-child links where the second-generation individual was aged 19–21 at the time of ACS response. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is whether the first-generation individual in each parent-child link is an adopted or step-parent. The outcome of interest in column (2) and column (3) is an indicator for whether the second-generation individual is attending college at the time of their response to the American Community Survey (2005–2019). Column (2) restricts the sample to parent-child links where the link captures a biological relationship. Column (3) restricts the sample to parent-child links where the link captures an adopted/step-child relationship. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year), first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table 4: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Labor Market Outcomes

	(1) log EARNINGS	(2) POSITIVE EARNINGS (\$)	(3) LFP	(4) UNEMPLOYED	(5) PUBLIC ASSISTANCE
Panel A: IV					
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.008 (0.005)	-185.7 (136.2)	0.0003 (0.0016)	0.0013 (0.0011)	0.0008 (0.0007)
Panel B: Reduced Form					
Nonattainment $\times$ Post	0.0087 (0.0058)	212 (151.9)	-0.0003 (0.0018)	-0.0015 (0.0013)	-0.0009 (0.0008)
Fixed Effects	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES
Observations	944,000	944,000	1,182,000	1,182,000	1,182,000
Control Mean	10.66	\$43,040	0.832	0.047	0.019
First Stage KP F-Stat	20.08	20.08	20.08	20.08	20.08

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. This sample of first-generation individuals is restricted to individuals that are aged 33–44 at the time of ACS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is the log of self-reported earnings. The outcome of interest in column (2) is self-reported earnings, restricted to those with positive earnings. The outcome of interest in column (3) is an indicator for whether an individual was participating in the labor force in the week prior to ACS response. The outcome of interest in column (4) is an indicator for whether the individual was unemployed in the week prior to ACS response. The outcome of interest in column (5) is an indicator for whether the individual was in receipt of any form of public assistance in the week prior to ACS response. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table 5: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Family Structure Outcomes

	(1) ANY LINKS	(2) NUMBER OF LINKS	(3) MAIN SAMPLE LINK	(4) AGE AT FIRST LINK	(5) MARRIED	(6) DIVORCED
Panel A: IV						
First-Gen Prenatal TSP ($10\mu g/m^3$)	0.0006 (0.0016)	0.0038 (0.0087)	0.0015 (0.0012)	-0.0014 (0.0305)	-0.0026 (0.0020)	0.0036** (0.0017)
Panel B: Reduced Form						
Nonattainment $\times$ Post	-0.0007 (0.0019)	-0.0048 (0.0108)	-0.0017 (0.0014)	0.0158 (0.0348)	0.0030 (0.0022)	-0.0042** (0.0017)
Fixed Effects	YES	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES	YES
Observations	1,633,000	1,633,000	1,182,000	1,182,000	1,182,000	1,182,000
Control Mean	0.81	1.93	0.058	26.15	0.77	0.13
First Stage KP F-Stat	18.55	18.55	20.08	20.08	20.08	20.08

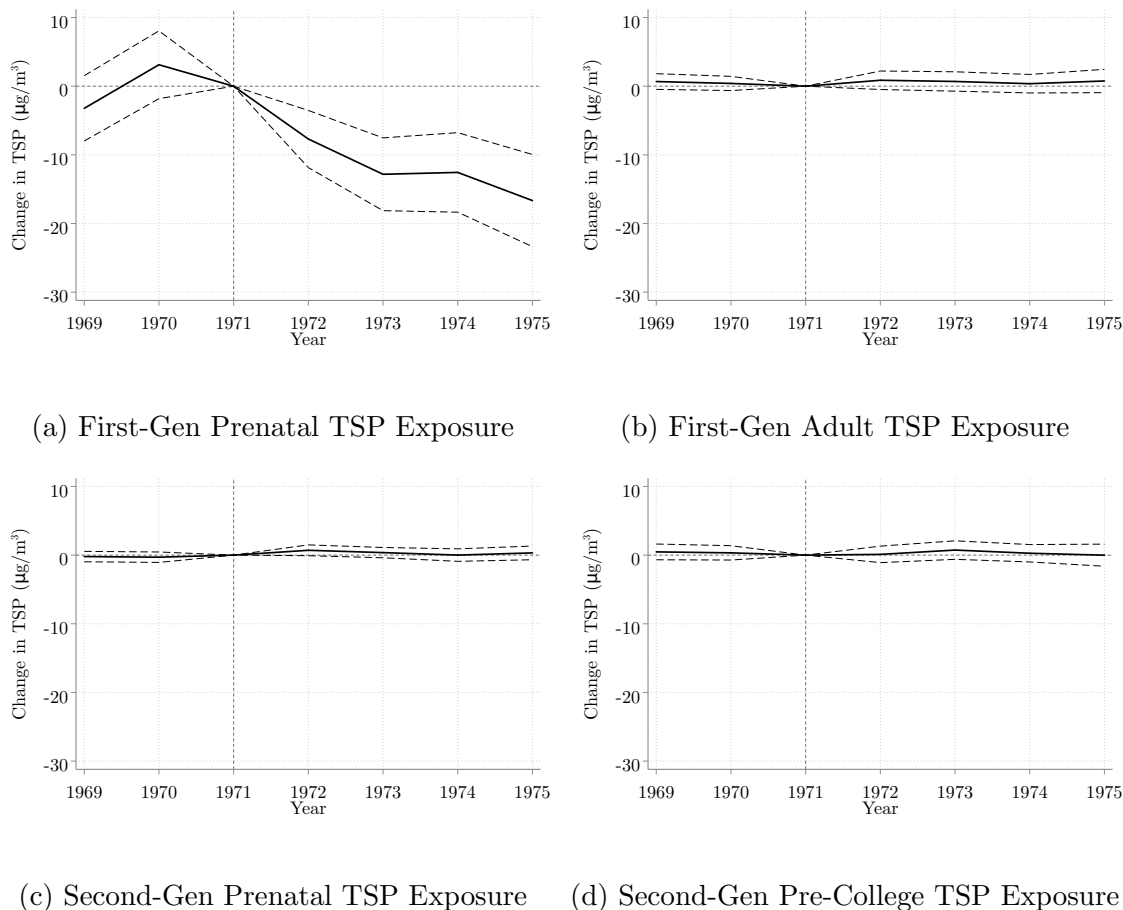
Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. The sample in columns (1) and (2) contains all first-generation individuals that are aged 33–44 at the time of ACS response. The sample in column (3)–(6) contains all first-generation individuals that are aged 33–44 at the time of ACS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is an indicator for whether the first-generation individual has a link to a second-generation individual. The outcome of interest in column (2) is a count of the number of second-generation links. The outcome of interest in column (3) is an indicator for whether any of the parent–child links are represented in our main second-generation analysis sample. The outcome of interest in column (4) is the age that the first-generation individual was when the first parent–child link was created. The outcome of interest in column (5) is an indicator for whether the individual is married. The outcome of interest in column (6) is an indicator for whether the individual is divorced. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. Standard errors are clustered at the first generation’s county of birth level. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table 6: The Relationship between First-Generation Prenatal TSP Exposure and parent–child Interactions

	(1) QUALITY TIME (minutes)	(2) HELPING CHILDREN (minutes)	(3) EATING MEALS (minutes)	(4) ARTS AND SPORTS (minutes)	(5) CHILD CARE (minutes)
Panel A: IV					
First-Gen Prenatal TSP ($10\mu g/m^3$)	-27.69** (13.02)	-5.35 (6.66)	-10.15* (6.05)	-11.96** (5.00)	1.03 (6.26)
Panel B: Reduced Form					
Nonattainment $\times$ Post	25.23** (10.61)	4.81 (5.99)	9.12** (4.55)	12.23*** (3.71)	-0.93 (5.57)
Fixed Effects	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES
Observations	5,600	5,600	5,600	5,600	5,600
Control Mean	118.60	31.75	41.30	7.85	37.73
First Stage KP F-Stat	12.51	12.51	12.51	12.51	12.51

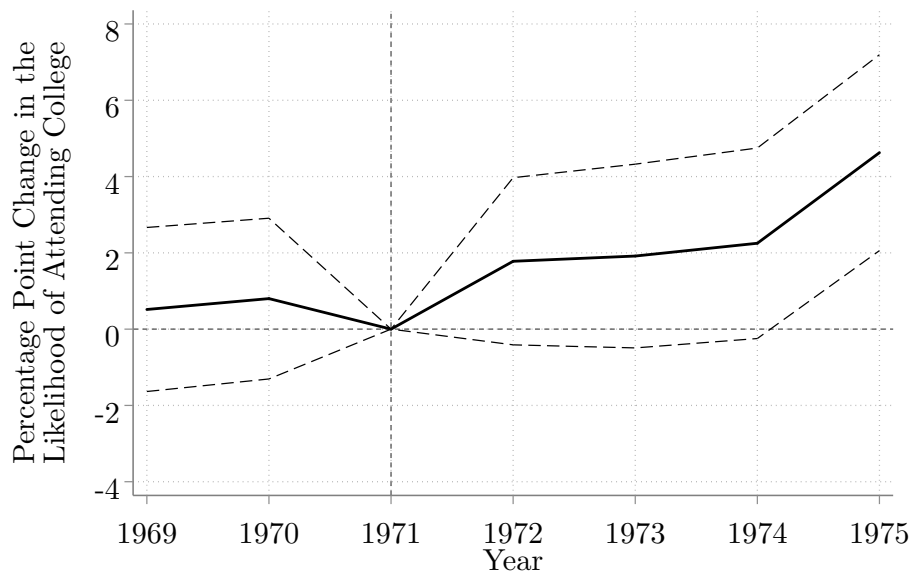
Source: Census Numident, Decennial Census 2000 and 2010, ACS 2005 through 2019, and American Time-Use Survey 2003–2019. Notes: The unit of analysis is a first-generation individual. The sample contains all first-generation individuals that are aged 25–45 at the time of ATUS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. Following [Price \(2008\)](#) our outcomes of interest are measures of “quality time”, which restricts time spent with children to include only activities in which the child was the primary focus of the activity, or where a reasonable amount of interaction between parent and child would likely arise. The outcome in column 1 is total quality time in minutes. The outcome in column 2 is the amount of time spent “reading to/with; playing, not sports; helping with homework; helping/teaching; talking with/listening to; arts and crafts”. The outcome in column 3 is the amount of time spent “eating meals together”. The outcome in column 4 is the amount of time spent “playing sports; attending performing arts; participating in religious activities”. The outcome in column 5 is the amount of time spent “looking after (as primary activity); physical care”. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Figure 1: The Relationship between First-Generation Prenatal Nonattainment Exposure and TSP Exposure



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and TSP exposure. The unit of analysis is a parent-child link for the sample of second-generation individuals that were aged 19–21 at the time of ACS response. Panel (a) presents estimates of the association between first-generation prenatal exposure to nonattainment and first-generation prenatal TSP exposure. This is the cohort-specific representation of the first stage of our instrumental variable analysis. Panel (b) presents estimates of the association between first-generation prenatal exposure to nonattainment and first-generation TSP exposure at the time of ACS response (aged 33–44). Panel (c) presents estimates of the association between first-generation prenatal exposure to nonattainment and second-generation gestational TSP exposure. Panel (d) presents estimates of the association between first-generation prenatal exposure to nonattainment and second-generation TSP exposure at age 16. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No other controls are included. Results are robust to the inclusion of first-generation prenatal controls as presented in Table 1. Standard errors are clustered at the first generation's county of birth level. Dashed lines represent 95% confidence intervals.

Figure 2: The Relationship between First-Generation Prenatal Nonattainment Exposure and Second-Generation College Attendance



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: This figure presents cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and second-generation college attendance. The unit of analysis is a parent-child link for the sample of second-generation individuals that were aged 19–21 at the time of ACS response. The estimates are conditional on first-generation county-of-birth, state-of-birth $\times$ year, birth-month, and year-of-birth $\times$ survey-year fixed effects (analogous to column (1) of Table 2). No other controls are included. Results are robust to including first-generation prenatal controls (analogous to column (2) of Table 2) and to including second-generation fixed effects and controls (analogous to column (3) of Table 2). Standard errors are clustered at the first generation's county of birth level. Dashed lines represent 95% confidence intervals.

Appendix Materials— For Online Publication

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A Additional Results and Robustness Tests

A.1 Descriptive Statistics

Table A1: Descriptive Statistics Prior to the 1970 CAAA

	(1) ATTAINMENT COUNTY	(2) NONATTAINMENT COUNTY	(3) DIFFERENCE (2)-(1)
Cohort Demographics			
Female	0.466	0.461	-0.005** (0.002)
White	0.772	0.732	-0.040 (0.026)
Black	0.073	0.103	0.030** (0.013)
Hispanic	0.063	0.085	0.023 (0.021)
Other	0.091	0.078	-0.013 (0.010)
Environmental Exposure			
Gestational Exposure ($\mu g/m^3$)	70.25	105.78	35.53*** (3.20)
Rainfall (days)	26.32	24.29	-2.03 (1.34)
Avg. Daily Maximum Temperature	17.87	17.89	0.02 (0.68)
County Socio-Economic Characteristics			
County Population	412,800	1,366,000	953,200** (425,300)
Personal Income per Capita (\$ 1969)	3,989	4,242.2	253.2* (134.0)

Source: Census Numident, Decennial Census 2000 and 2010, ACS 2005 through 2019, EPA, NOAA, BEA. Notes: These statistics are calculated using the main first-generation sample as in Table 4. We use all pre-treatment years (1969–1971) for the cohort demographics and environmental exposures. County socio-economic characteristics are for 1969. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01. Standard errors are clustered at the parent’s county of birth level.

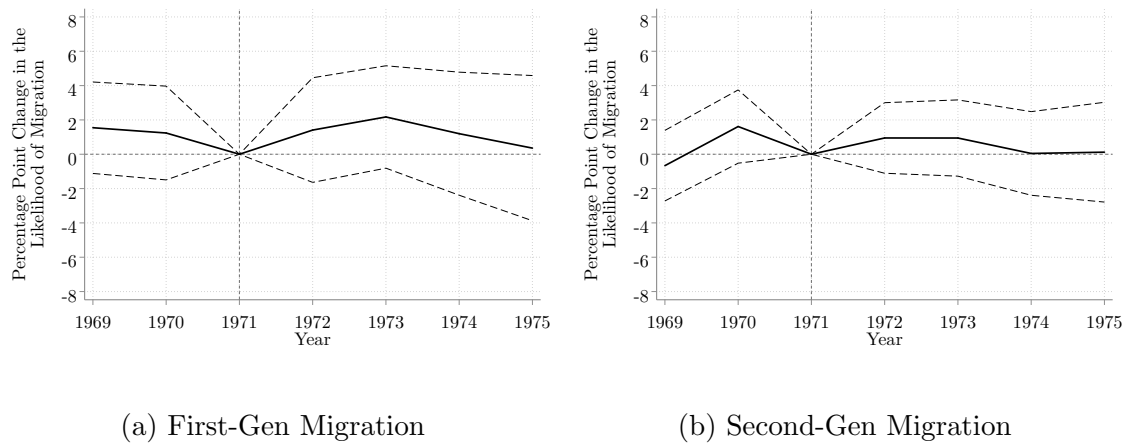
A.2 Migration

Table A2: The Relationship between First-Generation Prenatal TSP Exposure and Migration away from First-Generation County of Birth

	(1) FIRST-GEN MIGRATION (OUT OF COUNTY)	(2) SECOND-GEN MIGRATION (OUT OF COUNTY)
Panel A: IV		
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.015 (0.0099)	-0.0066 (0.0065)
Panel B: Reduced Form		
Nonattainment $\times$ Post	0.016 (0.010)	0.0075 (0.0070)
Fixed Effects	YES	YES
First-Gen Controls	YES	YES
Observations	87,500	139,000
Control Mean	0.627	0.572
First Stage KP F-Stat	13.05	13.93

Source: Census Numident, Decennial Census 2000 and 2010, ACS 2005 through 2019, and the Census MAFARF. Notes: The unit of analysis is a parent–child link. The analysis sample is parent–child links where the second-generation individual was aged 19–21 at the time of ACS response. The outcome in column (1) is an indicator for whether the parent lives in a different county at the time of ACS response to their county of birth. The outcome in column (2) is an indicator for whether the second-generation individual lives in a different county at age 16 to their parent’s county of birth. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). We also condition on first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the first-generation individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. The underlying sample size is constant across columns. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Figure A1: The Relationship between First-Generation Prenatal Nonattainment Exposure and Migration



Source: Census Numident, Decennial Census 2000 and 2010, ACS 2005 through 2019, and the Census MAFARF. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and the likelihood that first- and second-generation individuals live in a different county than the first-generation county-of-birth. The unit of analysis is a parent-child link. The analysis sample is parent-child links where the second-generation individual was aged 19–21 at the time of ACS response. Panel (a) presents estimates of the association between an indicator for whether the parent lives in a different county at the time of ACS response to their county of birth and first-generation prenatal exposure to nonattainment. Panel (b) presents estimates of the association between an indicator for whether the second-generation individual lives in a different county at age 16 to their parent’s county of birth and first-generation prenatal exposure to nonattainment. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No other controls are included. Results are robust to the inclusion of first-generation prenatal controls as presented in Table A2. Standard errors are clustered at the first generation’s county of birth level. Dashed lines represent 95% confidence intervals.

A.3 Maternal vs. Paternal Exposure

Table A3: The Relationship between First-Generation Prenatal TSP Exposure and Second-Generation College Attendance (Maternal vs. Paternal Exposure)

	SECOND-GENERATION COLLEGE ATTENDANCE		
	Mother Exposed (1)	Father Exposed (2)	Both Exposed (3)
Panel A: IV			
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0189** (0.00896)	-0.0189** (0.00952)	-0.0237** (0.01105)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.0215*** (0.0081)	0.0222** (0.0091)	0.0275*** (0.0100)
First-Gen Fixed Effects	YES	YES	YES
First-Gen Controls	YES	YES	YES
Observations	124,000	112,000	97,000
Control Mean	0.439	0.439	0.439
First Stage KP F-Stat	14.09	13.38	13.6

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a parent–child link. The analysis sample is parent–child links where the second-generation individual was aged 19–21 at the time of ACS response. The outcome of interest in all columns is an indicator for whether the second-generation individual is attending college at the time of their response to the American Community Survey (2005–2019). Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. All columns report estimates that condition on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year) as well as controls for first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the first-generation individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

A.4 Including Probabilistic Links

Table A4: The Relationship between First-Generation Prenatal TSP Exposure and Second-Generation College Attendance (Probabilistic Links)

	SECOND-GENERATION COLLEGE ATTENDANCE		
	(1)	(2)	(3)
Panel A: IV			
First-Gen Prenatal TSP ($10\mu\text{g}/\text{m}^3$)	-0.0132** (0.0056)	-0.0108* (0.0061)	-0.0093 (0.0057)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.0157*** (0.0056)	0.0126** (0.0060)	0.0108* (0.0059)
First-Gen Fixed Effects	YES	YES	YES
First-Gen Controls	NO	YES	YES
Second-Gen Controls	NO	NO	YES
Second-Gen Fixed Effects?	NO	NO	YES
Observations	219,000	219,000	219,000
Control Mean	0.455	0.455	0.455
First Stage KP F-Stat	20.93	14.49	15.00

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a parent–child link. The analysis sample is parent–child links where the second-generation individual was aged 19–21 at the time of ACS response. The sample is expanded to include probabilistic links as well as certain links. The outcome of interest in all columns is an indicator for whether the second-generation individual is attending college at the time of their response to the American Community Survey (2005–2019). Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. Column (1) reports estimates from the most parsimonious specification, which includes first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No additional controls are included. Column (2) reports estimates that additionally control for first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the first-generation individual’s nine month gestational period. Column (3) further includes second-generation fixed effects (county-of-birth, state-of-birth $\times$ year, and birth month) and second-generation demographic controls (indicators for sex, race/ethnicity, and age). The number of observations are rounded in concordance with US Census Bureau disclosure practices. The underlying sample size is constant across columns. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

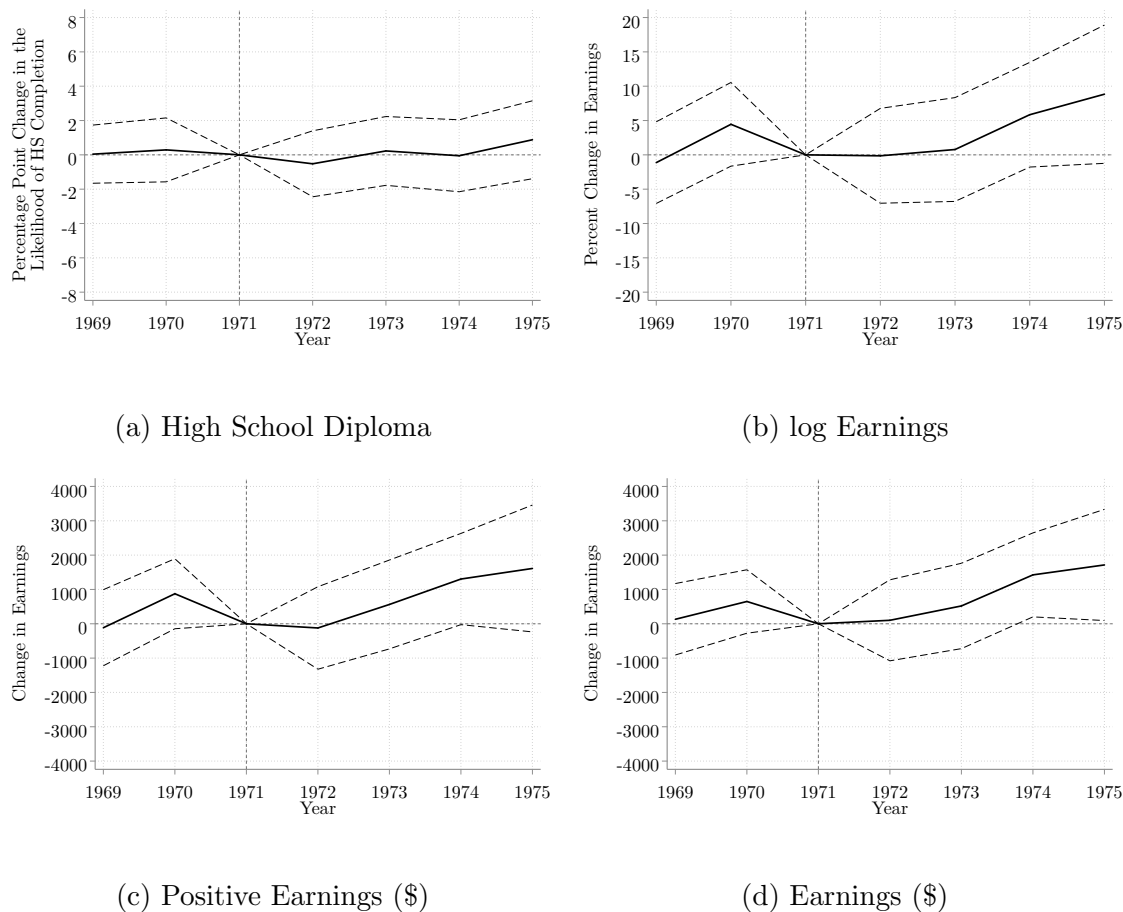
A.5 Additional Second-Generation Outcomes

Table A5: The Relationship between First-Generation Prenatal TSP Exposure on Second-Generation High School Completion and Earnings

	(1) HIGH SCHOOL DIPLOMA	(2) log EARNINGS	(3) POSITIVE EARNINGS (\$)	(4) EARNINGS (\$)
Panel A: IV				
First-Gen Prenatal TSP (10 μ g/m ³)	0.0012 (0.0043)	-0.010 (0.022)	-391.6 (403.9)	-640.7 (400.0)
Panel B: Reduced Form				
Nonattainment $\times$ Post	-0.0014 (0.0049)	0.010 (0.022)	387.8 (384.7)	627.1* (363.2)
First-Gen Fixed Effects	YES	YES	YES	YES
First-Gen Controls	YES	YES	YES	YES
Observations	139,000	78,000	78,000	98,500
Control Mean	0.838	10.16	25,940	20,590
First Stage KP F-Stat	20.17	15.68	15.68	16.06

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a parent-child link. In column 1 the analysis sample reflects parent-child links where the second-generation individual was aged 19–21 at the time of ACS response. In columns 2–4 the analysis sample reflects parent-child links where the second-generation individual was 23 or older at the time of ACS response. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is whether the the second-generation individual completed high school. The outcome of interest in column (2) is the log of self-reported earnings. The outcome of interest in column (3) is self-reported earnings, restricted to individuals with positive earnings. The outcome of interest in column (4) is self-reported earnings, including individuals with zero earnings. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year), first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual's nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation's county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Figure A2: The Relationship between First-Generation Prenatal Nonattainment Exposure and Additional Second-Generation Economic Outcomes



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019.
Notes: The unit of analysis is a parent-child link. In panel (a) the analysis sample reflects parent-child links where the second-generation individual was aged 19–21 at the time of ACS response. In panels (b)–(d) the analysis sample reflects parent-child links where the second-generation individual was 23 or older at the time of ACS response. The outcome of interest in panel (a) is whether the the second-generation individual completed high school. The outcome of interest in panel (b) is the log of self-reported earnings. The outcome of interest in panel (c) is self-reported earnings, restricted to individuals with positive earnings. The outcome of interest in panel (d) is self-reported earnings, including individuals with zero earnings. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No other controls are included. Results are robust to the inclusion of first-generation prenatal controls as presented in Table A5. Standard errors are clustered at the first generation’s county of birth level. Dashed lines represent 95% confidence intervals.

B Exploring Mechanisms

What are the mechanisms through which prenatal pollution exposure propagates across generations? We posit that there are two broad channels through which these effects could arise. The first channel is biological. Gestational pollution exposure could result in epigenetic changes; i.e., permanent changes in gene expression. If so, hereditary transmission of these changes could affect the second generation. The second mechanism is economic. Gestational pollution exposure affects the human capital and earnings of the first generation. In turn, parental resources and investments may affect their children’s human capital accumulation.

B.1 Conceptual Framework

To consider the mechanisms through which prenatal exposure to pollution could directly affect individuals and consequently have intergenerational effects, we extend the framework presented by [Isen et al. \(2017\)](#).³⁵

Let an individual’s health stock be a function of inputs during two time periods: $h = h(p_1, p_2, X(p_1))$, where p_1 represents prenatal pollution exposure, p_2 represents post-birth pollution exposure, and X reflects their genetic endowment, which may be affected by prenatal pollution exposure through epigenetic effects (permanent changes in gene expression). In our analysis the treated cohort born after the 1970 CAAA is exposed to lower p_1 and p_2 . By contrast, cohorts born in these counties before the CAAA came into effect had higher p_1 but the same exposure to p_2 (assuming they continued to live in their counties of birth). Under the assumption that the policy does not induce differential changes in later-life pollution exposure, this comparison isolates the additional effect of changes in prenatal pollution exposure p_1 .

B.1.1 First-Generation Effects

For the first generation – those directly affected – we model the effects of prenatal pollution on earnings, which is a function of the health stock, h , and education, e ,

$$y_f = y_f(e_f, h_f) = y_f(e_f(h_f(p_{f,1}, p_{f,2}, X(p_{f,1}))), h_f(p_{f,1}, p_{f,2}, X(p_{f,1})))$$

where y_f represents earnings and e_f represents years of schooling. The channels through which prenatal pollution exposure affects earnings can be characterized by taking the total derivative of earnings, y_f , with respect to prenatal pollution exposure $p_{f,1}$,

$$\frac{dy_f}{dp_{f,1}} = \frac{\partial y_f}{\partial e_f} \frac{\partial e_f}{\partial h_f} \left[\frac{\partial h_f}{\partial p_{f,1}} + \frac{\partial h_f}{\partial X} \frac{\partial X}{\partial p_{f,1}} \right] + \frac{\partial y_f}{\partial h_f} \left[\frac{\partial h_f}{\partial p_{f,1}} + \frac{\partial h_f}{\partial X} \frac{\partial X}{\partial p_{f,1}} \right]$$

Prenatal pollution exposure, p_1 , affects the health stock, h , which affects earnings directly, $\frac{\partial y}{\partial h}$, as well as indirectly through educational attainment, $\frac{\partial y}{\partial e} \frac{\partial e}{\partial h}$.

³⁵The foundations of this model are based on earlier work by [Grossman \(1972\)](#), [Bleakley \(2010\)](#), and [Cunha and Heckman \(2007\)](#).

B.1.2 Second-Generation Effects

For the second generation, we model the effects of first-generation (parental) prenatal pollution exposure on second-generation educational attainment, which is a function of the second generation's health stock, h_s . The health stock of the second generation is a function of second-generation prenatal pollution exposure, $p_{s,1}$, second-generation later-life pollution exposure, $p_{s,2}$, first-generation earnings, $y_f(h_f, e_f)$, first-generation education, $e_f(h_f)$, first-generation health stock, $h_f(p_{f,1}, p_{f,2}, X(p_{f,1}))$, and any genetic changes that may have occurred due to first-generation prenatal pollution exposure or second-generation prenatal pollution exposure, $X(p_{f,1}, p_{s,1})$,

$$e_s(h_s(p_{s,1}, p_{s,2}, X(p_{f,1}, p_{s,1})); y_f(h_f(p_{f,1}, p_{f,2}, X(p_{f,1})), e_f(h_f(\cdot)); e_f(h_f(\cdot)); h_f(\cdot)))$$

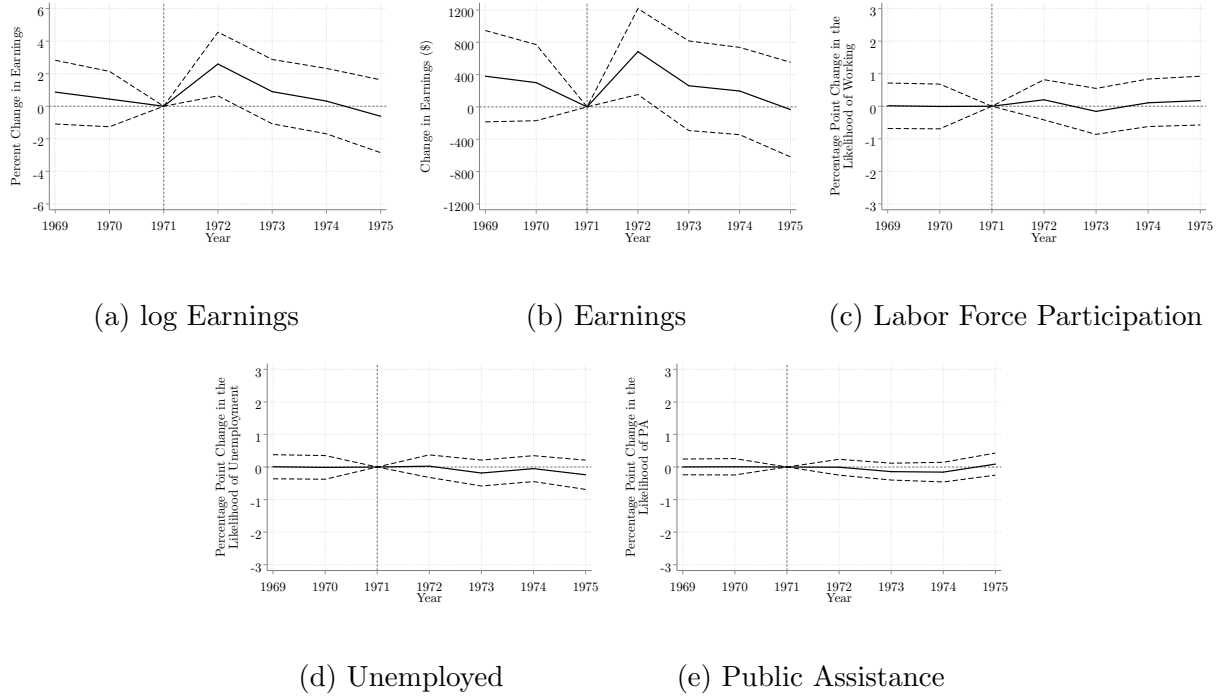
The channels through which prenatal pollution exposure affects earnings can be characterized by taking the total derivative of educational attainment, e_s , with respect to parental prenatal pollution exposure, $p_{f,1}$,

$$\begin{aligned} \frac{de_s}{dp_{f,1}} &= \frac{\partial e_s}{\partial h_s} \frac{\partial h_s}{\partial X} \frac{\partial X}{\partial p_{f,1}} && \text{(Biological Effects)} \\ &+ \frac{\partial e_s}{\partial y_f} \frac{\partial y_f}{\partial h_f} \left[\frac{\partial h_f}{\partial p_{f,1}} + \frac{\partial h_f}{\partial X} \frac{\partial X}{\partial p_{f,1}} \right] + \frac{\partial e_s}{\partial y_f} \frac{\partial y_f}{\partial e_f} \frac{\partial e_f}{\partial h_f} \left[\frac{\partial h_f}{\partial p_{f,1}} + \frac{\partial h_f}{\partial X} \frac{\partial X}{\partial p_{f,1}} \right] && \text{(Income Effects)} \\ &+ \frac{\partial e_s}{\partial e_f} \frac{\partial e_f}{\partial h_{f,1}} \left[\frac{\partial h_{f,1}}{\partial p_{f,1}} + \frac{\partial h_f}{\partial X} \frac{\partial X}{\partial p_{f,1}} \right] && \text{(Parental Education)} \\ &+ \frac{\partial e_s}{\partial h_f} \left[\frac{\partial h_f}{\partial p_{f,1}} + \frac{\partial h_f}{\partial X} \frac{\partial X}{\partial p_{f,1}} \right] && \text{(Parental Health)} \end{aligned}$$

Broadly speaking, parental prenatal pollution exposure could affect the educational attainment of the second generation through either biological channels, $\frac{\partial e_s}{\partial h_s}$, or the household environment, i.e, parental resources, $\frac{\partial e_s}{\partial y_f}$, and human capital, $\frac{\partial e_s}{\partial e_f}$ and $\frac{\partial e_s}{\partial h_f}$.

B.2 Cohort-Specific Estimates of First-Generation Economic Outcomes

Figure B1: The Relationship between First-Generation Prenatal Nonattainment Exposure and First-Generation Economic Outcomes



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and first-generation economic outcomes. The unit of analysis is a first-generation individual. The sample of first-generation individuals is restricted to individuals that are aged 33-44 at the time of ACS response and have at least one parent-child link. The outcome of interest in panel (a) is the log of self-reported earnings at the time of ACS response. The outcome of interest in panel (b) is self-reported earnings, restricted to those with positive earnings. The outcome of interest in panel (c) is an indicator for labor force participation in the week prior to ACS response. The outcome of interest in panel (d) is an indicator for being unemployed in the week prior to ACS response. The outcome of interest in panel (e) is an indicator for being in receipt of any form of public assistance in the week prior to ACS response. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No other controls are included. Results are robust to the inclusion of first-generation prenatal controls as presented in Table 4. Standard errors are clustered at the first generation's county of birth level. Dashed lines represent 95% confidence intervals.

B.3 First-Generation Educational Outcomes

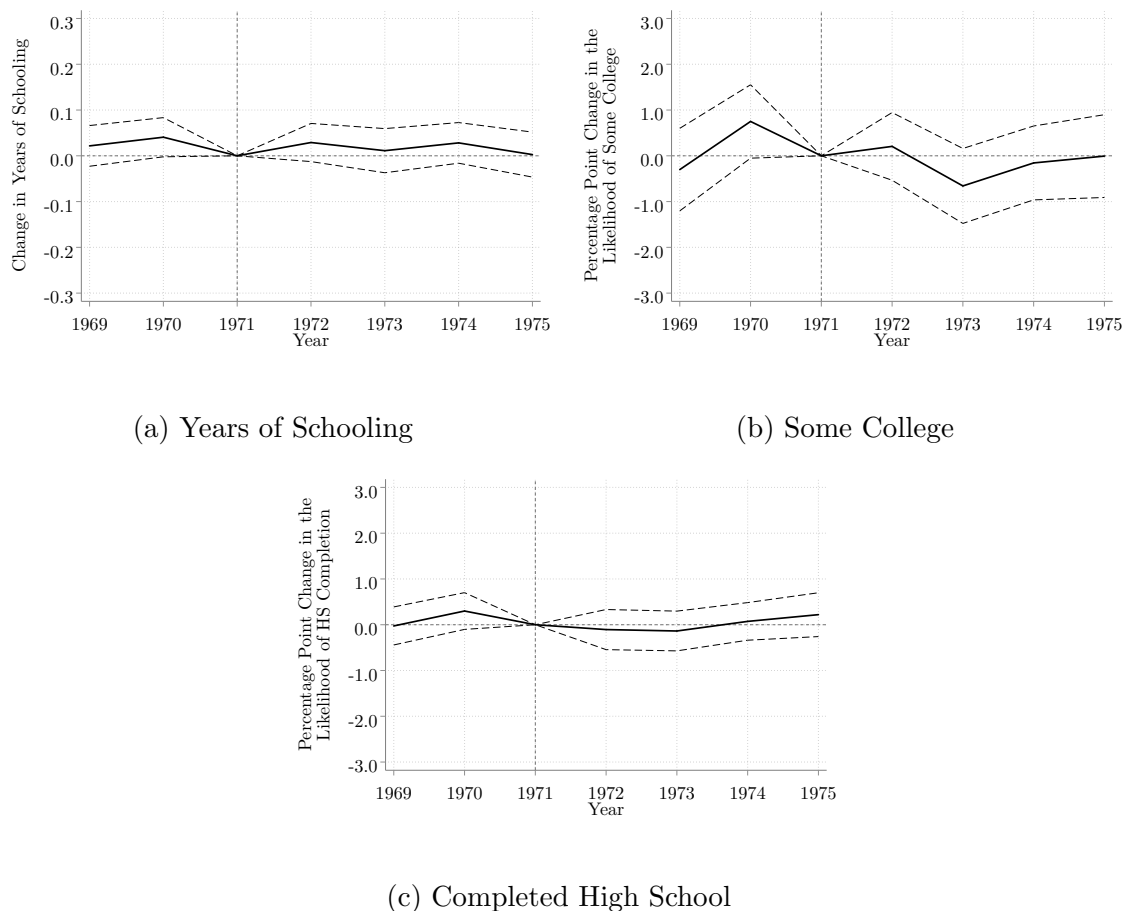
Table B1: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Educational Outcomes

	(1) YEARS OF SCHOOLING	(2) SOME COLLEGE	(3) COMPLETED HIGH SCHOOL
Panel A: IV			
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0159 (0.0125)	0.0015 (0.0025)	0.0005 (0.0012)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.0183 (0.0137)	-0.0017 (0.0028)	-0.0006 (0.0014)
Fixed Effects	YES	YES	YES
Individual Controls	YES	YES	YES
County-level Controls	YES	YES	YES
Observations	1,182,000	1,182,000	1,182,000
Control Mean	14.73	0.689	0.942
First Stage KP F-Stat	20.08	20.08	20.08

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. This sample of first-generation individuals is restricted to individuals that are aged 33-44 at the time of ACS response and have at least one parent-child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is the number of years of schooling. The outcome of interest in column (2) is an indicator for whether the individual completed at least some coursework at a college or university. The outcome of interest in column (3) is an indicator for whether the individual completed high school. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). In addition, we include controls for first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual's nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. The underlying sample size is constant across columns. Standard errors are clustered at the first generation's county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

B.4 Cohort-Specific Estimates of First-Generation Educational Outcomes

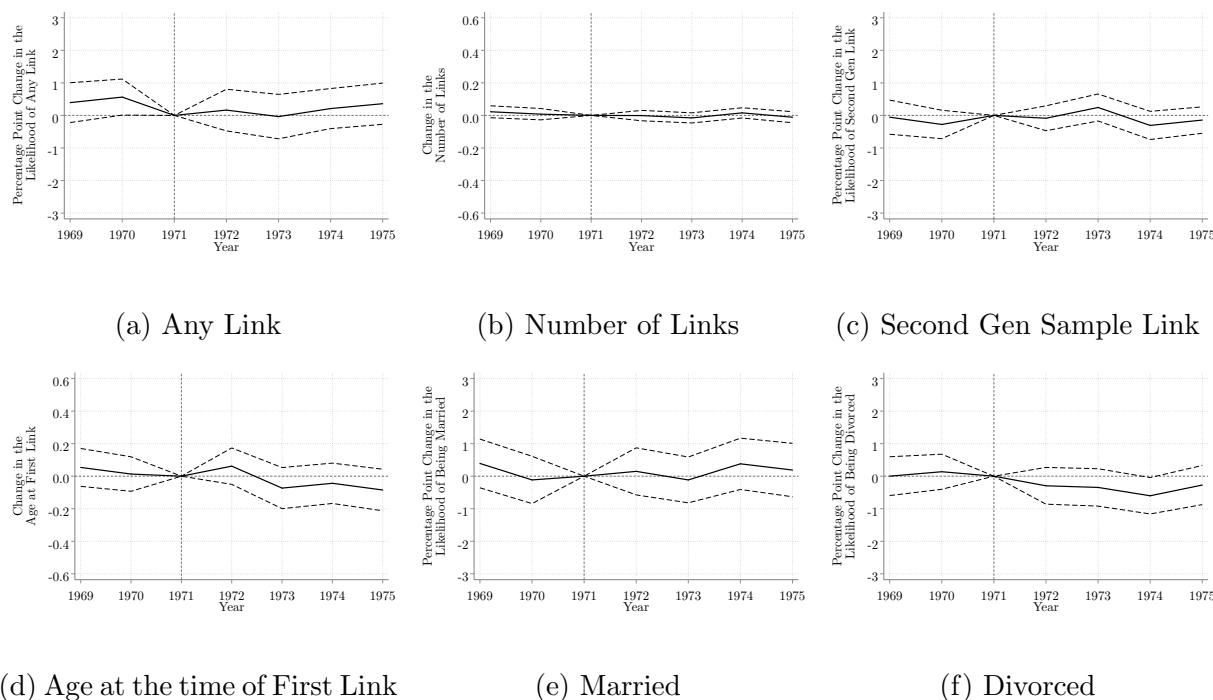
Figure B2: The Relationship between First-Generation Prenatal Nonattainment Exposure and First-Generation Educational Outcomes



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and first-generation educational outcomes. The unit of analysis is a first-generation individual. The sample of first-generation individuals is restricted to individuals that are aged 33–44 at the time of ACS response and have at least one parent–child link. The outcome of interest in panel (a) is the number of years of completed schooling. The outcome of interest in panel (b) is an indicator for whether the individual completed some college. The outcome of interest in panel (c) is an indicator for whether the individual finished high school. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No other controls are included. Results are robust to the inclusion of first-generation prenatal controls as presented in Table B1. Standard errors are clustered at the first generation’s county of birth level. Dashed lines represent 95% confidence intervals.

B.5 Cohort-Specific Estimates of First-Generation Family Structure Outcomes

Figure B3: The Relationship between First-Generation Prenatal Nonattainment Exposure and First-Generation Family Structure



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and first-generation family structure outcomes. The unit of analysis is a first-generation individual. The sample in panels (a) and (b) contains all first-generation individuals that are aged 33–44 at the time of ACS response. The sample in panels (c)–(f) contains all first-generation individuals that are aged 33–44 at the time of ACS response and have at least one parent–child link. The outcome of interest in panel (a) is an indicator for whether the first-generation individual has a link to a second-generation individual. The outcome of interest in panel (b) is a count of the number of second-generation links. The outcome of interest in panel (c) is an indicator for whether any of the parent–child links are represented in our main second-generation analysis sample. The outcome of interest in panel (d) is the age that the first-generation individual was when the first parent–child link was created. The outcome of interest in panel (e) is an indicator for whether the individual is married. The outcome of interest in panel (f) is an indicator for whether the individual is divorced. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). No other controls are included. Results are robust to the inclusion of first-generation prenatal controls as presented in Table 5. Standard errors are clustered at the first generation’s county of birth level. Dashed lines represent 95% confidence intervals.

B.6 Economic Outcomes by Sex

Table B2: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Labor Market Outcomes (Women)

	(1) log EARNINGS	(2) POSITIVE EARNINGS (\$)	(3) LFP	(4) UNEMPLOYED	(5) PUBLIC ASSISTANCE
Panel A: IV					
First-Gen Prenatal TSP (10 μ g/m ³)	-0.0039 (0.0059)	-47.05 (195.7)	-0.0003 (0.0019)	0.0024 (0.0016)	-0.0000 (0.0008)
Panel B: Reduced Form					
Nonattainment $\times$ Post	0.0045 (0.0068)	54.40 (226.4)	0.0004 (0.0022)	-0.0027 (0.0018)	0.0000 (0.0009)
Fixed Effects	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES
Observations	448,000	448,000	511,000	511,000	511,000
Control Mean	10.86	\$52,320	0.926	0.046	0.011
First Stage KP F-Stat	18.42	18.42	18.42	18.42	18.42

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. This sample of first-generation individuals is restricted to women that are aged 33–44 at the time of ACS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is the log of self-reported earnings. The outcome of interest in column (2) is self-reported earnings, restricted to those with positive earnings. The outcome of interest in column (3) is an indicator for whether an individual was participating in the labor force in the week prior to ACS response. The outcome of interest in column (4) is an indicator for whether the individual was unemployed in the week prior to ACS response. The outcome of interest in column (5) is an indicator for whether the individual was in receipt of any form of public assistance in the week prior to ACS response. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, and birth month fixed effects. In addition, we include controls for race/ethnicity, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table B3: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Labor Market Outcomes (Men)

	(1) log EARNINGS	(2) POSITIVE EARNINGS (\$)	(3) LFP	(4) UNEMPLOYED	(5) PUBLIC ASSISTANCE
Panel A: IV					
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0105 (0.0081)	-303.8 (193.9)	0.0006 (0.0024)	0.0005 (0.0015)	0.0015 (0.0011)
Panel B: Reduced Form					
Nonattainment $\times$ Post	0.0119 (0.0091)	342.8 (211.6)	-0.0008 (0.0027)	-0.0006 (0.0017)	-0.0017 (0.0012)
Fixed Effects	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES
Observations	496,000	496,000	671,000	671,000	671,000
Control Mean	10.45	\$34,550	0.759	0.048	0.027
First Stage KP F-Stat	21.72	21.72	21.53	21.53	21.53

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. This sample of first-generation individuals is restricted to men that are aged 33–44 at the time of ACS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is the log of self-reported earnings. The outcome of interest in column (2) is self-reported earnings, restricted to those with positive earnings. The outcome of interest in column (3) is an indicator for whether an individual was participating in the labor force in the week prior to ACS response. The outcome of interest in column (4) is an indicator for whether the individual was unemployed in the week prior to ACS response. The outcome of interest in column (5) is an indicator for whether the individual was in receipt of any form of public assistance in the week prior to ACS response. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, and birth month fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

B.7 Educational Outcomes by Sex

Table B4: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Educational Outcomes (Women)

	(1) YEARS OF SCHOOLING	(2) SOME COLLEGE	(3) COMPLETED HIGH SCHOOL
Panel A: IV			
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0047 (0.0018)	0.0045 (0.0040)	0.0009 (0.0019)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.0055 (0.0201)	-0.0052 (0.0043)	-0.0011 (0.0022)
Fixed Effects	YES	YES	YES
Individual Controls	YES	YES	YES
County-level Controls	YES	YES	YES
Observations	511,000	511,000	511,000
Control Mean	14.58	0.651	0.937
First Stage KP F-Stat	18.42	18.42	18.42

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019.
Notes: The unit of analysis is a first-generation individual. This sample of first-generation individuals is restricted to women that are aged 33-44 at the time of ACS response and have at least one parent-child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is the number of years of schooling. The outcome of interest in column (2) is an indicator for whether the individual completed at least some coursework at a college or university. The outcome of interest in column (3) is an indicator for whether the individual completed high school. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). In addition, we include controls for first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual's nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation's county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table B5: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Educational Outcomes (Men)

	(1) YEARS OF SCHOOLING	(2) SOME COLLEGE	(3) COMPLETED HIGH SCHOOL
Panel A: IV			
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.0241 (0.0156)	-0.0009 (0.0029)	0.0002 (0.0015)
Panel B: Reduced Form			
Nonattainment $\times$ Post	0.0274 (0.0171)	0.0010 (0.0032)	-0.0002 (0.0017)
Fixed Effects	YES	YES	YES
Individual Controls	YES	YES	YES
County-level Controls	YES	YES	YES
Observations	672,000	672,000	672,000
Control Mean	14.86	0.723	0.948
First Stage KP F-Stat	21.52	21.52	21.52

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019.
Notes: The unit of analysis is a first-generation individual. This sample of first-generation individuals is restricted to men that are aged 33-44 at the time of ACS response and have at least one parent-child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is the number of years of schooling. The outcome of interest in column (2) is an indicator for whether the individual completed at least some coursework at a college or university. The outcome of interest in column (3) is an indicator for whether the individual completed high school. All estimates are conditional on first-generation fixed effects (county-of-birth, state-of-birth $\times$ year-of-birth, month-of-birth, and year-of-birth $\times$ survey-year). In addition, we include controls for first-generation race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual's nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation's county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

B.8 Family Structure by Sex

Table B6: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Family Structure Outcomes (Women)

	(1) ANY LINKS	(2) NUMBER OF LINKS	(3) MAIN SAMPLE LINK	(4) AGE AT FIRST LINK	(5) MARRIED	(6) DIVORCED
Panel A: IV						
First-Gen Prenatal TSP (10 μ g/m ³)	-0.0007 (0.0020)	0.0005 (0.0115)	0.0044*** (0.0002)	0.0033 (0.0045)	-0.0010 (0.0029)	0.0007 (0.0023)
Panel B: Reduced Form						
Nonattainment $\times$ Post	0.0008 (0.0025)	-0.0006 (0.0142)	- 0.0051*** (0.0016)	-0.0382 (0.0505)	0.0012 (0.0034)	-0.0008 (0.0026)
Fixed Effects	YES	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES	YES
Observations	852,000	852,000	511,000	511,000	511,000	511,000
Control Mean	0.852	2.108	0.043	27.1	0.811	0.110
First Stage KP F-Stat	19.1	19.1	18.42	18.42	18.42	18.42

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. The sample in columns (1) and (2) contains all first-generation women that are aged 33–44 at the time of ACS response. The sample in columns (3)–(6) contains all first-generation women that are aged 33–44 at the time of ACS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is an indicator for whether the first-generation individual has a link to a second-generation individual. The outcome of interest in column (2) is a count of the number of second-generation links. The outcome of interest in column (3) is an indicator for whether any of the parent–child links are represented in our main second-generation analysis sample. The outcome of interest in column (4) is the age that the first-generation individual was when the first parent–child link was created. The outcome of interest in column (5) is an indicator for whether the individual is married. The outcome of interest in column (6) is an indicator for whether the individual is divorced. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

Table B7: The Relationship between First-Generation Prenatal TSP Exposure and First-Generation Family Structure Outcomes (Men)

	(1) ANY LINKS	(2) NUMBER OF LINKS	(3) MAIN SAMPLE LINK	(4) AGE AT FIRST LINK	(5) MARRIED	(6) DIVORCED
Panel A: IV						
First-Gen Prenatal TSP ($10\mu g/m^3$)	0.0021 (0.0024)	0.0084 (0.0131)	-0.0010 (0.0016)	-0.0471 (0.0440)	-0.0035 (0.0028)	0.0056** (0.0024)
Panel B: Reduced Form						
Nonattainment $\times$ Post	-0.0026 (0.0029)	-0.0104 (0.0160)	0.0011 (0.0019)	0.0535 (0.0477)	0.0040 (0.0031)	- 0.0063*** (0.0024)
Fixed Effects	YES	YES	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES	YES	YES
Observations	782,000	782,000	671,000	671,000	671,000	671,000
Control Mean	0.769	1.726	0.070	25.41	0.732	0.150
First Stage KP F-Stat	17.86	17.86	21.53	21.53	21.53	21.53

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. The sample in columns (1) and (2) contains all first-generation men that are aged 33–44 at the time of ACS response. The sample in columns (3)–(6) contains all first-generation men that are aged 33–44 at the time of ACS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is an indicator for whether the first-generation individual has a link to a second-generation individual. The outcome of interest in column (2) is a count of the number of second-generation links. The outcome of interest in column (3) is an indicator for whether any of the parent–child links are represented in our main second-generation analysis sample. The outcome of interest in column (4) is the age that the first-generation individual was when the first parent–child link was created. The outcome of interest in column (5) is an indicator for whether the individual is married. The outcome of interest in column (6) is an indicator for whether the individual is divorced. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

B.9 Assortative Matching

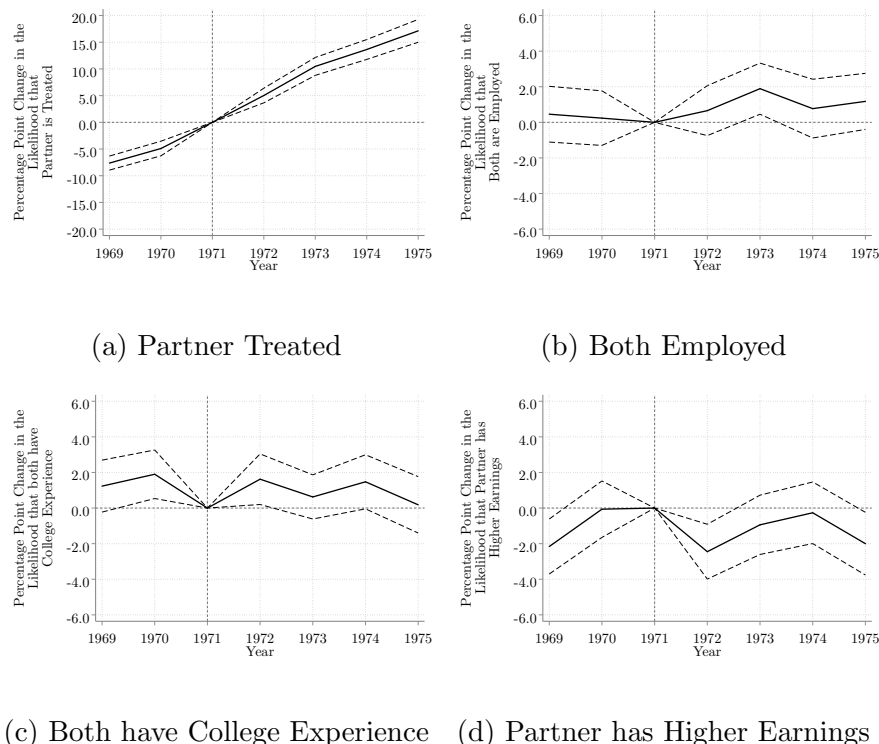
Table B8: The Relationship between First-Generation and Partner Characteristics

	(1) PARTNER TREATED	(2) BOTH EMPLOYED	(3) BOTH COLLEGE	(4) PARTNER HIGHER INCOME
Panel A: IV				
First-Gen Prenatal TSP ($10\mu g/m^3$)	-0.1159*** (0.0274)	-0.0066 (0.0041)	-0.0035 (0.0037)	0.0031 (0.0039)
Panel B: Reduced Form				
Nonattainment $\times$ Post	0.1369*** (0.0063)	0.0078* (0.0044)	0.0042 (0.0045)	-0.0036 (0.0046)
Fixed Effects	YES	YES	YES	YES
Individual Controls	YES	YES	YES	YES
County-level Controls	YES	YES	YES	YES
Observations	284,000	284,000	284,000	284,000
Control Mean	0.207	0.678	0.680	0.233
First Stage KP F-Stat	18.66	18.66	18.66	18.66

Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: The unit of analysis is a first-generation individual. The sample is restricted to all first-generation individuals that are married, aged 33–44 at the time of ACS response, and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. The outcome of interest in column (1) is an indicator for whether the first-generation individual’s partner was affected by the treatment. The outcome of interest in column (2) is an indicator equal to one if both partners are employed, and zero otherwise. The outcome of interest in column (3) is an indicator equal to one if both partners are college educated and zero otherwise. The outcome of interest in column (4) is an indicator equal to one if the first-generation individual’s partner has higher earnings. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. In addition, we include controls for race/ethnicity and sex, pre-1970 CAAA county of birth economic characteristics (employment, total transfer income, personal income per capita) interacted with quadratic time trends, and county of birth temperature and precipitation measured during the individual’s nine month gestational period. The number of observations are rounded in concordance with US Census Bureau disclosure practices. Standard errors are clustered at the first generation’s county of birth level. Significance levels are indicated as * 0.10 ** 0.05 *** 0.01.

B.10 Cohort-Specific Estimates of Assortative Matching

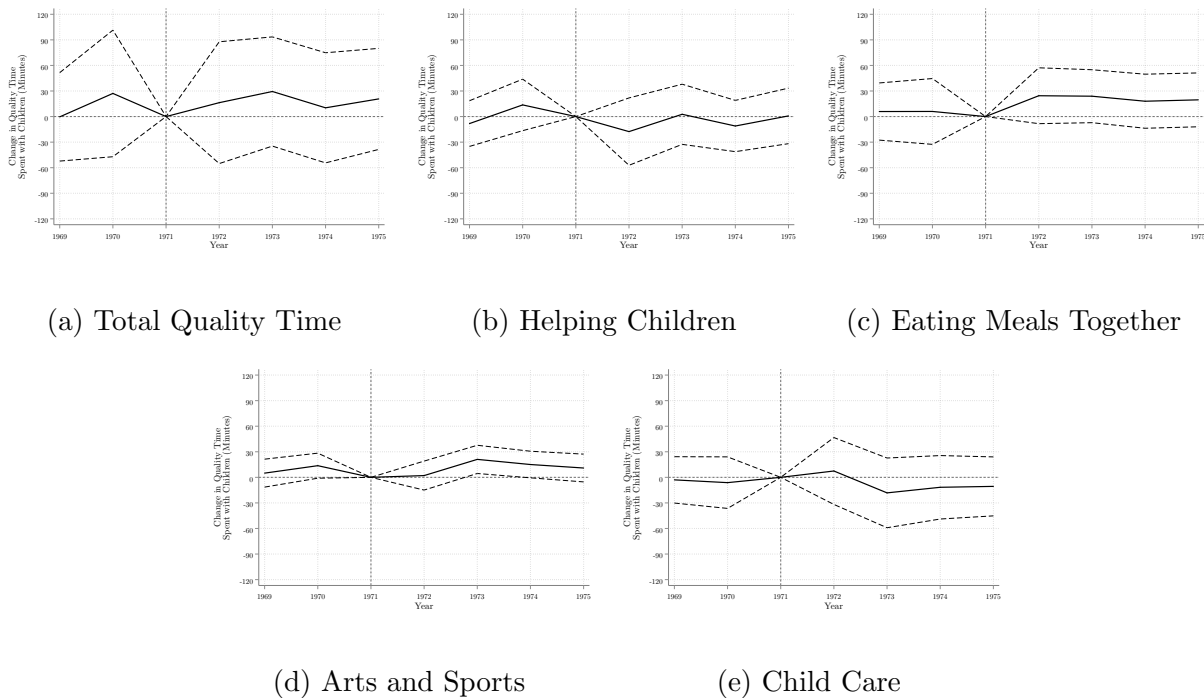
Figure B4: The Relationship between First-Generation Prenatal Nonattainment Exposure and Partner Characteristics



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and parent–child interactions. The sample is restricted to all first-generation individuals that are married, aged 33–44 at the time of ACS response, and have at least one parent–child link. The outcome of interest in panel (a) is an indicator for whether the first-generation individual’s partner was affected by the treatment. The outcome of interest in panel (b) is an indicator equal to one if both partners are employed, and zero otherwise. The outcome of interest in panel (c) is an indicator equal to one if both partners are college educated and zero otherwise. The outcome of interest in panel (d) is an indicator equal to one if the first-generation individual’s partner has higher earnings. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. No other controls are included. Results are robust to including individual-level and county-level controls as presented in Table B8. Standard errors are clustered at the first generation’s county of birth level. Dashed lines represent 95% confidence intervals.

B.11 Cohort-Specific Estimates of Time-Use Outcomes

Figure B5: The Relationship between First-Generation Prenatal Nonattainment Exposure and Parent–Child Interactions



Source: Census Numident, Decennial Census 2000 and 2010 and ACS 2005 through 2019. Notes: These figures present cohort-specific estimates of the association between first-generation prenatal exposure to nonattainment and parent-child interactions. The unit of analysis is a first-generation individual. The sample in column contains all first-generation individuals that are aged 25–45 at the time of ATUS response and have at least one parent–child link. Panel A reports instrumental variable estimates. Panel B reports reduced form estimates. Following [Price \(2008\)](#) our outcomes of interest are measures of “quality time”, which restricts time spent with children to only include activities in which the child was the primary focus of the activity, or where a reasonable amount of interaction between parent and child would likely arise. The outcome in panel (a) is total quality time in minutes. The outcome in panel (b) is the amount of time spent “reading to/with; playing, not sports; helping with homework; helping/teaching; talking with/listening to; arts and crafts”. The outcome in panel (c) is the amount of time spent “eating meals together”. The outcome in panel (d) is the amount of time spent “playing sports; attending performing arts; participating in religious activities”. The outcome in panel (e) is the amount of time spent “looking after (as primary activity); physical care”. All estimates are conditional on an individual’s county-of-birth, state-of-birth $\times$ year, birth month, and year-of-birth $\times$ survey-year fixed effects. No other controls are included. Results are robust to including individual-level and county-level controls as presented in Table 6. Standard errors are clustered at the first generation’s county of birth level. Dashed lines represent 95% confidence intervals.

C Data Appendix

C.1 The Census Bureau’s Data Linkage Infrastructure

This analysis heavily relies on the Census Bureau’s linkage infrastructure, which allows researchers inside the Census Bureau firewall to link data across surveys, Decennial Census data and administrative records using probabilistically assign linkage keys called Protected Identification Keys (PIKs). For more information on the linkage processes that assign these PIKs and the linkage infrastructure overall, see [Wagner and Layne \(2014\)](#).

C.2 Parent–Child Links

To study the intergenerational effects of the Clean Air Act, it is necessary to locate parents at birth (around the enactment of 1970 CAAA), link these parents to their children, and measure outcomes for both parents and children. We begin by assembling a database of all parent–child links that can be evaluated using the various data sources available in the Census Data Linkage Infrastructure. The set of links we are able to identify is not, we should stress, the full population of links.

To benchmark our link coverage, consider that the completed cohort fertility rate for women born in 1970 is about 2.1. There were about 44 million women aged 30–50 in the 2010 Census (i.e., born between 1960 and 1980). Taking the 1970 CCFR as constant throughout this group, we can expect at most 92 million biological children. In practice, we identify fewer than this, due to linkage error, and the fact that women who are born in the latter part of our birth year range have not completed fertility in the latest available data used to identify parent–child links (the 2015 ACS).

C.2.1 Decennial Census Data

The 2000 and 2010 Decennial Census 100 percent detail file (HDF), colloquially the “Census short form”, collects an abbreviated set of demographic information from the full population of the United States in decadal Census years. This demographic information includes date of birth, sex, race and ethnicity, and some household structure information. Unfortunately, the relationship information collected in the Census does not capture the full relationship structure within a household. Rather, the Census collects information from each individual in a household on their relationship to the householder.

This means it is possible to identify two types of parent child links: “certain” parent–child links between a child and the householder parent, and “probable” parent–child links between a child and the married or unmarried partner of their parent householder. The relationship codes are sufficiently detailed to separate biological children of a householder, adopted children, and stepchildren. For the purposes of the project at hand, we identify only parent–child links (certain or probable) for parents born between 1960–1980.

To identify these two types of links in the 2000 Census HDF, we use the following algorithm. We first subset the HDF by age and relationship code, retaining only individuals aged 40 or younger (i.e. who were born after 1960) who are householders, the spouse or unmarried partner of a householder, biological child of householder, adopted child of householder, or

the stepchild of householder. Then, for each household, we assign three link variables: “Certain Parent”, which is the PIK of the householder, “Probable-Married”, which is the PIK of the householder’s spouse, and “Probable-Unmarried”, which is the PIK of the householder’s unmarried partner. Each of these variables are missing if the relevant PIK is missing (due to PIK non-assignment when the HDF was analyzed via PVS). We then reshape the data into long form (so each row contains the child’s PIK, the parent’s PIK, and indicators for the type of child and the type of parent). We discard all cases where the child or parent’s PIKs are missing.

This yields a dataset containing about 65 million parent–child links. Of these, about 35 million are “Certain” links, about 28 million are “Probable-Married”, and the remaining approximately 2 million are “Probable-Unmarried”. We identify more mother-child links ($\approx$ 38 million) than father-child links (27 million). As expected, the parent–child links identified in the 2000 HDF are heavily tilted toward the older parents: about 51 million links involve parents born before 1970, while about 14 million involve parents born after 1980.

We repeat the use of the same algorithm to identify parent–child links in the 2010 HDF. We identify substantially more links in the 2010 Census, as expected. In all, we identify 115 million parent–child links—of these, about 64 million are “Certain”, 46 million are “Probable-Married”, and the remaining 5 million are “Probable-Unmarried”. As with the 2000 HDF, we identify more mother-child links (65 million) compared to father-child links (50 million). We continue to identify more parent–child links for parents born before 1970, although the split is much more even compared to the 2000 HDF (reflecting the fact that women born before 1970 had largely completed fertility, while women born after 1970 were still in prime childbearing age ranges).

Combining the information from the two Decennial Census files, we can identify about 152 million unique parent–child links for about 81 million children. Note that because of the way that the “Probable” links are identified, it is possible that some of these links represent changes in family structure (marriages, divorces, and creation/dissolution of unmarried partnerships). About 123 million links occur for children with 1 or 2 unique links, while the remaining 29 million occur for children with 3 or more links (these represent cases where parental relationships appear to have changed).

C.2.2 Other Demographic Surveys

The final source of data on parent–child links comes from demographic surveys. These surveys are substantially smaller than the Census, but allow us to identify relationships in non-Decennial year. We use the American Community Survey, which has been conducted monthly since 2001, as our main survey source. The Current Population Survey Annual Social and Economic Supplement (CPS ASEC), which is conducted every March, is another potential source of parent–child links. However, the CPS ASEC sample size is substantially smaller than the ACS (200,000 individuals in the CPS-ASEC versus about 5 million in the ACS 1 year files) and thus would provide minimal additional information relative to the Decennial Census and ACS.

From 2005–2015, the ACS has consisted of a sample size of about 5 million individuals. The content of the ACS has evolved considerably over this period. In particular, the household relationship question was redesigned for the 2008 ACS. From 2001–2007, the ACS

relationship question allowed for 10 categories, with a single “child of householder” category that includes adopted children, biological children and step-children. From 2008–2015, the relationship question was expanded to 13 categories, with separate categories for adopted children, biological children and step-children. As with the Decennial HDF data, the relationship variable in the ACS contains categories for married and unmarried partners of the householder, so we categorize the ACS links as “Certain” (for the householder), “Probable-Married” (for the householder’s spouse), and “Probable-Unmarried” (for the householder’s unmarried partner). We are able to collect about 22 million parent–child links for about 12 million children.

C.2.3 Combining Relationship Information

The relationship information we have extracted from Decennial Census data and demographic surveys has a substantial degree of overlap. In total, we identify over 168 million links.

Note while a vast majority (about 87 percent) of children can be linked to one or two parents, there are a substantial number who are linked to three or more parents.

Although we retain links from each source, to allow for robustness checks on the type of link used, our main estimating sample uses only the certain links derived from the Census and ACS data. Some source information is included in the data, including the parent and child types from the Census and survey data and the year(s) a link appears in the ACS data. In robustness exercises, we additionally include probabilistic links.

C.3 Pollution Exposure at Birth

To analyze the intergenerational effects of pollution exposure, we need to be able to infer the level of ambient air pollution and the changes in EPA policy (designation nonattainment of NAAQS) that parents were exposed to at birth. We do this in three steps. First, we link the set of unique parents identified in the previous section to the Census Numident to obtain date and place of birth. We then obtain monitor-level daily pollution measures from the EPA, which aggregate to the county level, and link these county-level measures to the parents’ place of birth. Finally, since the EPA’s records of nonattainment designations appear to be incomplete or destroyed, we simulate these nonattainment designations for counties with EPA monitors active in 1969 (before CAAA 1970).

C.3.1 Census Numident Data

Our source of information on the parents’ place of birth comes from the Census Numident, which is a derivative product of the SSA Numerical Identification File. The Census Numident contains three fields that can be used to infer place of birth, which are transcribed from form SS-5 (application for social security number). First, there is a flag for country of birth, which contains a two-digit code for the country of birth for non-native born individuals. Second, there is a state-of-birth field, which contains a two-character abbreviation for state of birth for all native born US citizens. Both of these fields can be assigned one-to-one with standard geographies (i.e. FIPS codes). The third relevant field, containing information on place of

birth, however, is slightly more cumbersome. This variable represents the first 12 characters of the place (or county) of birth entered on form SS-5. There is little standardization or cleaning done by SSA or Census for this field, and thus, there are numerous misspellings and inconsistencies.

In order to match the information in the *pobcity* with standardized geographies (i.e. county FIPS codes), we take a two-step approach. First, after excluding foreign-born individuals (about 13 million parents), we capitalize on a crosswalk developed jointly by Census researchers and external researchers (described in detail in (Isen et al., 2017)). This crosswalk provides all exact matches (after standardization) and probabilistic matches between place of birth entries and unique GNIS place names. A second crosswalk between GNIS places and county FIPS codes allows us to directly match parents to counties exactly. For the remaining cases, we execute a probabilistic matching algorithm. This algorithm assigns a match by calculating the optimal string alignment (OSA) distance between a place of birth entry and a reference list of all county and Census place names, selecting the smallest distance (maximum of 5) within state of birth. This is essentially the same algorithm as in Voorheis (2017). All told, about 74 percent of native-born parents can be assigned a place of birth using the GNIS crosswalks, and another 23 percent can be matched using our probabilistic matching algorithm, so that about 97 percent of native born parents can be assigned a county of birth.

C.3.2 EPA Monitor Data

With information about the place of parents’ birth in hand, we can infer the level of pollution exposure experienced by these individuals if we have some information based on the average exposure within their county of birth.

To gather this pollution exposure information, we rely on monitor data from the EPA. The EPA has made monitor-level air quality data available via the AQDM API. Our pollutant of interest is particulate matter. For the relevant period of time (around 1970), the primary regulated pollutant was total suspended particles (TSP), defined as the density of particulates less than 100 microns, measured in units $\mu g/m^3$.³⁶ We thus retrieve all TSP (EPA pollutant code 11101) monitor observations between 1969–1980.

The TSP standard was set based on a 24-hour sampling, and hence, the monitor-level data is provided on a daily basis. Our baseline approach to aggregating these daily monitor-level observations is as follows. For each county-day, we calculate the average TSP concentration across all active monitors in that day, which we take as the average exposure to TSP in that county on that day. We then calculate county-level moving average exposure to TSP for each unique birthday between 1969 and 1980 for two periods of interest: the nine months before birth (in utero exposure) and the year after birth (infant exposure).

The EPA’s monitoring network expanded dramatically following the passage of CAAA 1970, expanding both the number of counties monitored and the density of monitors within consistently monitored counties. This poses two potential challenges to our baseline measurement approach above. First, some counties will only have observations in the “post-treatment” period in our OLS and IV regressions. Second, even for counties which are

³⁶This definition was later revised to 10 microns (PM10) and 2.5 microns (PM2.5) standards in 1987 and 1997 respectively.

consistently monitored, the expansion of the monitor network may result in systematic measurement error—average county exposure will be more precisely measured with more monitors and so the pre-treatment observations are more likely to be mismeasured than the post-treatment observations. To address these issues, we also produce county-level moving averages using a constant set of monitors (the monitors that were active in 1969 or earlier).

C.3.3 Nonattainment Designations

Our empirical strategy relies on identifying the intergenerational effects of pollution exposure at birth using plausibly exogenous variation in TSP exposure that resulted from the EPA designating counties as in nonattainment of the ambient air quality standards in the 1970 CAAA. Although the EPA made nonattainment designations publicly available starting in 1991, and researchers have reconstructed nonattainment designations back to 1980, there appear to be no existing records on which counties were initially designated as being in nonattainment in 1972, the first year in which the 1970 CAAA was in effect. This is because the EPA designated entire air quality regions as in nonattainment, not individual counties, when setting standards authorized under CAAA 1970. In practice, however, the way that these nonattainment designations were applied bound at the county level: the EPA considered an air quality district to be in nonattainment if any of the monitors in the region were in nonattainment. Thus, in practice, counties with monitors in attainment were not subject to the same regulations as counties that were in nonattainment. Since the TSP air quality standards are known, and as noted in the previous section, we have monitor-level data on the actual level of exposure in the years before the 1970 CAAA was in effect, it is possible to reconstruct which counties would have been in nonattainment.

Nonattainment of the primary air quality standard for TSP set in 1970 CAAA occurs if either a) the annual average (geometric mean) TSP concentration is above $75 \mu g/m^3$, or b) the second highest daily TSP concentration is above $260 \mu g/m^3$. We use the monitor-level observations from the previous section to calculate the geometric mean and second highest daily TSP concentration for all counties with at least one monitor in 1970. This allows us to categorize 258 counties as “nonattainment” counties and 319 counties as “attainment” counties.

C.3.4 Other County Attributes

Estimating the effects of pollution exposure at birth on adult outcomes for parents and intergenerational effects for their children may be confounded by other characteristics of the parents’ place of birth, such as weather or economic activity. To this end, we obtain pre-determined (i.e. before the clean air act of 1970) information on county level economic activity from the BEA and county-level weather information from the National Oceanic and Atmospheric Administration (NOAA).

Following [Isen et al. \(2017\)](#), we obtain information on the economy and population of U.S. counties in 1969 from the Bureau of Economic Analysis’ Regional Economic Accounts (1969 is the earliest year for which the BEA publishes regional accounts data). We extract four variables of interest from the regional accounts: total population, total employment, total personal income, and total personal transfer income. From these, we can construct income

per capita and employment-to-population ratio measures; these measures allow us to control for important county-level economic characteristics that may confound the nonattainment-pollution relationship.

Additionally, we obtain information on county-level weather patterns. Temperature and precipitation, in particular, play important factors in the formation of particulate matter emissions and in the suspension of particulate matter in the atmosphere after emission. Importantly, there is evidence both that very low temperatures can increase PM concentrations by emissions (at cold temperatures, internal combustion engines burn fuel less efficiently), while very high temperatures can increase PM concentrations through suspension and atmospheric particle formation (sulfate and nitrate particles form more readily at hot temperatures). Additionally, precipitation decreases PM concentrations by decreasing suspension. Thus, we obtain weather-station level data on daily high temperature, low temperature, and precipitation from NOAA’s Global Historical Climatology Network (GHCN). For each day between 1959 and 1981, we interpolate across the weather station network to each county centroid using inverse distance weighting to obtain a county-day level dataset. We can then calculate the average high/low temperature and number of precipitation days corresponding to the 9 months before birth and the year after birth for each individual.

C.4 ATUS Data

To investigate mechanisms underlying the second generation effect, we will leverage a secondary linked dataset which will allow us to measure both time use for individuals at a point in time, as well as their place of birth and the level of pollution they were exposed to. We do this by linking a subset of respondents to the American Time Use Survey (ATUS) to the Census Numident.

Using the public use ATUS data from 2003–2019, we build a series of time use variables, which divide the total time spent during the reference day on specific child-enrichment activities (time spent on children’s education activities, time spent on children’s health activities, time spent reading to a child), as well as broad categories of non-sleep time use (time spent on work, time spent on social activities, time spent on leisure, time spent on education). We then link a subset of the ATUS respondents to the Census Numident to attach place of birth characteristics as follows.

Our linkage strategy relies on the fact that the ATUS sample frame is drawn from the Current Population Survey. Hence, it is possible to link ATUS respondents to the CPS on an individual level in the public use data. For the subset of individuals who are in sample and respond to the ASEC, we can link this public use identifier to the internal confidential CPS-ASEC data. The internal CPS-ASEC has had PIKs assigned, so we are then able to link this subset of individuals to the Census Numident by PIK, identifying place of birth and TSP exposure at birth using the same method used for the ACS sample as described above. We further subset this linked sample to individuals born between 1969 and 1980, coinciding with the first generation for the main ACS results. Note that this is a relatively small subsample of ATUS respondents (the final analysis sample for parents aged 25-45 has about 5,600 observations).